Finding Buyers: Information and Trade Promotion in the First Globalization

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November 3, 2026

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Abstract

Information frictions impede the formation of buyer–supplier linkages and are particularly severe across national borders. Because information about foreign buyers is costly to acquire privately but can be shared at negligible marginal cost, governments fund export promotion agencies to supply it. This paper estimates how much providing publicly available information about foreign buyers increases firms' export market participation. Identification comes from the administratively predetermined rotation of buyer listings published by 19th-century Austro-Hungarian consular offices. They covered only a few countries per issue for reasons of space constraints rather than commercial priority. The setting isolates the information channel since consular listings were close to the only public signal firms received about buyers abroad. I construct a novel panel linking 5,500 Austro-Hungarian exporting firms over 1890–1898 to decompose which features of public buyer information drive exports. Publication alone raises the probability of entering a destination by 5.7% over the baseline of 4.9%. When the listed buyers are predominantly intermediaries, the effect reaches 18.3% and entrants are more likely to survive past the first year. Buyer counts matter as much as buyer identity for entering new markets.

Keywords: buyer-supplier links, export promotion, search frictions, industrial policy, economic history

JEL Codes: F14, N73, O14, L25

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1 Introduction

Information frictions can be a substantial barrier to the formation of firm-to-firm linkages (Allen, 2014), and they are likely to bind even more tightly in international trade (Eaton et al., 2026). Governments commonly address these frictions through export promotion agencies, which assist firms in entering and expanding in foreign markets. Yet we know little about whether public intervention is effective in reducing cross-border information frictions for two reasons: first, studying export promotion bundles (Volpe Martincus and Carballo, 2008; Buus et al., 2025a) make it difficult to evaluate the effect of information friction reduction programs next to other policy components, such as fixed cost subsidies. Second, the literature on specific components with well-identified methods, such as RCTs (Breinlich et al., 2017; Ali et al., 2025) is thin, and their results often differ especially on the extensive margin from non-experimental papers. This is important since governmental spending can be substantial[1] and is often justified with the goal of assisting firms with foreign market entrance. At the intersection of cross-border information frictions and export promotion literature, this paper studies the causal effect of foreign buyer information provision on export market participation. By observing provided information, I aim to uncover the mechanism behind firms' utilization of learning about whom to contact in a foreign market.

I use a historical setting from the last 19th-century Austria-Hungary. As commercial interests expanded beyond the well-connected markets, Austria-Hungary invested heavily in trade-facilitating institutions from the 1870s onward. Consular offices became a key component on trade facilitation, and from 1890, they were explicitly tasked to collect information on local importing firms, on top of providing general market intelligence. The Ministry of Commerce consolidated the results into a widely available registry — the *Export Compass* — published annually from 1890 to 1898, then biennially until 1915. The setting generates quasi-experimental variation in buyer information availability. Each year, consular offices, in an administratively predetermined set of destination countries, were instructed to compile lists of locally active importers. This staggered, rotating coverage across six regions (from the Balkans to South America) provides quasi-random variation to observe changes in export activity of Austro-Hungarian firms.

To measure the effect of buyer information on export decisions, I construct a firm-year-destination country panel using the editions of the Export Compass from 1890 to 1898, covering 5,500 firms. A firm receives a signal[2] in given year from a certain destination country if the Export Compass lists at least one buyer from the destination country that imports a product that is exported by the firm. For instance, a wine producer in Austria-Hungary receives a signal from Romania in 1890

[1]For example, in the US in 2024 the Department of Commerce requested $379m for export promotion services (U.S. Department of Commerce, 2024)

[2]Throughout this paper, the term *signal* refers to the arrival of buyer information from a destination country in a given year, as published in the Export Compass. This differs from the standard use of the term in information economics, where signals are strategic actions taken by informed parties to communicate private information (see e.g. Spence, 2002). Here, the signal is an exogenous information shock received by the exporter from a third-party institution.

if there is at least one Romanian wine importer that appears in the registry that year. Then, I measure whether the observed Austro-Hungarian wine producer exports wine to Romania in 1891 and beyond.

The main specification shows a 0.287 percentage point increase compared to a baseline probability of 4.9%, which corresponds to a 5.9% increase. As expected, strong signals—defined either above the mean of median number of potential buyers—correspond to an increase in probability by 0.874 percentage points, corresponding to a 17.8% increase. The placebo tests show no pre-trends or anticipatory effects: future signals do not predict future export status. The only significant result comes with a year delay, which is consistent with the information publication of the Export Compass: since the buyer information and firms' export destinations were published in the same editions, it should not be possible to observe firm's reaction to the buyer signals in the same year. Furthermore, establishing new buyer-supplier linkages were unlikely within one year due to slower communication in the 19th century.

The effect does not seem to persist beyond one year. Next year exports to a certain destination country is only affected by information from a year before, however, signals from two or three years before are not statistically significant. The analysis is conducted both assuming a signal that switches on and off only in one year, or assuming signals to stay on for another one or two years, capturing the possibility that firms may have used a version of the Export Compass from previous years. The signal is significant only on the 10% level with a moderate magnitude if we assume the signal is turned on for two years, consistent with diluting the effect of baseline specification. The other results are not distinguishable statistically from zero.

I then separate entry into new destinations from continuation in markets a firm already serves. The pooled effect turns out to be driven mainly by the latter: interacting the signal with past export status to the same destination, firms already serving a market respond by about 4.2 percentage points, while firms not yet serving it show no measurable response, so the pooled estimate is this large effect diluted across a sample in which incumbent firm–destination pairs are a small minority. Restricting to observations with no prior export to the destination—93.6 percent of the sample—a signal as such does not raise the probability of entry. It does once the listing is thick enough: when the Compass names at least five matched buyers, entry rises significantly, by roughly a third of the pooled effect. Buyer information thus worked along two distinct margins. A single listed buyer was enough to sustain or renew an existing commercial relationship, whereas breaking into an untried market required a set of prospects large enough to justify the fixed costs of entry—evidence that what firms needed was not merely a contact but a market worth approaching.

A natural concern is that buyer signals proxy for product-specific demand booms rather than information: firms may read a long buyer list as evidence that a market wants their goods, instead of using the buyer information per se. I address this using a second component of the *Export-Compass*, the *Specielle Artikel* sections, in which consular posts reported on individual commodities. From 426 such entries I code whether the consul was recommending entry or expansion — identified

grammatically, through the Konjunktiv II mood in which such recommendations were invariably written, rather than through sentiment — and whether Austro-Hungarian producers were already established in that commodity. Signals accompanied by an encouraging report, and signals from markets where Austro-Hungarian goods were already established, do elicit stronger responses; but signals carrying neither feature remain significant and close to the headline estimate, so the effect does not seem to be driven by contemporaneous demand conditions.

Across destination markets, the response is concentrated where information is novel and cheapest to act on: signals from Mediterranean destinations, readily served from the ports of Trieste and Fiume, roughly double the baseline effect, while point estimates for distant overseas markets are close to zero—suggesting that the actionability of information is more important than novelty, or at least learning about buyers in remote destination alone is insufficient to move firms into novel markets.

Amongst potential buyers I observe whether a buyer is a wholesaler intermediary, defined as importers who buy goods in large quantities and redistribute them in their local markets. A listed wholesale buyer opens a different entry pathway than a direct buyer, and may be more likely to aggregate demand across multiple domestic suppliers (Ahn et al., 2011). This distinction turns out to be quantitatively central: signals dominated by wholesale intermediaries raise the probability of export entry roughly four times more than other signals if the list of buyers contain more than 50% of wholesale buyers. Furthermore, having a high share of wholesaler intermediaries in a signal results in longer-lasting buyer-supplier relations: signals with over 50% of wholesaler share two or three years before still predict a higher export presence a year later.

I then turn to which firms were able to use the information. Since every firm in the sample already exports, the question is not whether buyer lists bring new firms into foreign trade but which exporters can act on information about an unfamiliar market. One might expect the listings to matter most for narrowly specialized firms, which have the least private access to contacts abroad; the evidence points the other way. The response is concentrated among firms with prior export breadth: the signal effect is 1.27 percentage points larger for firms serving many destinations and 0.86 points larger for firms exporting many product lines, and the baseline coefficient becomes indistinguishable from zero once these interactions are included. Public information therefore did not substitute for export capability but complemented it. Knowing *whom* to approach is of limited use when the costs of serving a particular market remain, and these are destination-specific: experience in one market does not pay the cost of entering another, but it lowers it. The listings moved firms that had already built the capacity to extend their reach, not those for whom each new market was a first step.

Finally, I aggregate the analysis to the firm level to ask whether buyer information expanded firms' export portfolios or merely redirected them. The results in this section are correlational and suggestive since the identification relies on firm-country-year variation. Two patterns emerge. First, the geographic scope of exporting responds to the depth of information rather than its

breadth: the total number of matched buyers a firm could identify across markets raises its number of export destinations in the following year, while the mere count of signaled destinations does not. Because total destination counts rise with buyer information, the destination-level entry effects do not reflect pure reallocation across markets—the Compass expanded exporting on net rather than shifting it from uncovered to covered destinations. Second, firms receiving signals from more destinations narrow their exported product range, consistent with specialization: identified foreign demand allowed firms to concentrate on the product lines for which buyers were known to exist. Together, the firm-level results suggest that public buyer information changed not only where firms sold, but also what they sold.

This project contributes to three strands of literature. First, a growing literature documents that search and matching frictions between buyers and suppliers are a first-order impediment to trade, comparable in magnitude to physical transport costs (Allen, 2014; Bernard and Moxnes, 2018; Lenoir et al., 2023; Eaton et al., 2022; Miyauchi, 2024). In firms' search efforts, information frictions are particularly important (Jensen, 2007; Allen, 2014; Jensen and Miller, 2018). Causal evidence confirms that reducing these frictions has economically significant consequences: the elimination of information lags raises trade volumes and reduces price dispersion (Steinwender, 2018); experimental reductions in search costs increase trade flows and producer revenues (Bergquist et al., 2024); and improved buyer–seller matching raises product quality and firm profits (Atkin et al., 2017; Cai et al., 2024). I contribute to this literature by focusing on cross-border information frictions: First, because I observe the exact content of the information firms received (e.g. commercial function), I can decompose which attributes of a buyer signal generate new firm-to-firm linkages in international trade. Second, I analyze whether public information provided by governmental services are effective in reducing such firm-to-firm cross-border information frictions.

Second, export promotion policies have been widely evaluated across countries (Volpe Martincus et al., 2010; Benguria, 2021; Broocks and Van Biesebroeck, 2017; Buus et al., 2025b; Van Biesebroeck et al., 2015; Hiller, 2012; Görg et al., 2008; Schminke and Van Biesebroeck, 2013; Kim and Kim, 2026), with non-experimental studies—typically based on matching or difference-in-differences designs—finding mostly positive effects on both the intensive and extensive margins of exporting (with Bernard and Jensen (2004) a notable exception). Randomized evaluations paint a less optimistic picture: Breinlich et al. (2017); Kim et al. (2018); Ali et al. (2025); González et al. (2025) find no effect on the extensive margin. Two explanations could reconcile the discrepancy: unobserved confounders inflating the non-experimental estimates, or that specific interventions in the RCTs are short-lived or do not target the right friction. The treatments studied are consistent with the latter: Breinlich et al. (2017), Cusolito et al. (2023), and Kim et al. (2018) evaluate one-off provision of general information about export practices rather than sustained, actionable intelligence about specific counterparties, and Ali et al. (2025), who study a substantial subsidy covering up to 50% of fixed entry costs such as trade-fair participation and foreign representation, find effects only on exports to already-established destinations. My setting speaks directly to this divide. The

Export Compass combined features that neither side of the literature observes together: quasi-experimental variation from the rotation schedule, which addresses the confounding concern in observational studies, and a treatment that was specifically designed responding to firms' demand for actionable information. The positive extensive-margin effects I find suggest that the weaker results in the aforementioned RCTs may reflect the nature of the assistance provided: generic export training does not tell a firm whom to sell to; a buyer list does.

Third, evidence on industrial policies, including policies reducing information frictions in the 19th century (such as Steinwender (2018)), are rare, due to both the lack of micro-data on firms' export activity and the scarcity of natural experiments that empowers researchers to identify the effectivity of policy tools. The analysis of industrial policies are especially difficult not only due to endogeneity issues, but also due to the fact that policies are never implemented randomly by real policy makers (Juhász et al., 2024). Reduced form evidence on trade protection shows a consistent positive effect of technology adoption (Juhász, 2018) and firm outcomes (Hanlon, 2020), however, these papers use semi-random variation resulting from exogenous events rather than from the implementation of policy tools. This project, similar to Békés et al. (2026), analyses a policy aimed at assisting firms fostering business connections by reducing information frictions on several domain. While Békés et al. (2026) analyses a large-scale world fair in Paris in 1900, the policy in this project has a different approach: information is provided each year from different angles of the known world, thus firms have the opportunity to learn about the markets and potential buyers through several years, choosing an export strategy that fits their needs and competencies best. To my best knowledge, there are no other papers analyzing export promotion (or information services) that use 19th century micro data. However, other mechanisms are explored using well-identified settings, such as trademark protection in Alfaro et al. (2022), intellectual property protection in Moser (2005, 2012), domestic market integration in Donaldson and Hornbeck (2016); Donaldson (2018), or the effects of colonialism in Romero (2023).

The paper is structured as follows. Section 2 describes the institutional context of trade in the first globalization: falling trade barriers, the transportation and communication revolutions, and the state-provided information services—above all the consular network—that emerged to facilitate firms entering into international markets. Section 3 presents the data sources and the construction of the firm-level panel, including the LLM-assisted parsing and product matching that assigns treatment, along with descriptive statistics on signals and export probabilities. Section 4 lays out the empirical strategy: the quasi-random rotation of destination coverage, the baseline specification, the confounders the design absorbs, and the remaining threats to identification. Section 5 presents the main results and reports robustness checks, including placebo signals, sample restrictions, and a continuous treatment measure. Section 6 traces out mechanisms through heterogeneity in signal strength, destination geography, and buyer composition. Section 7 aggregates the analysis to the firm level to examine effects on firms' overall export portfolios. Section 8 summarizes the results and connects them to policy discussions.

2 Trade institutions in the first globalization

By the end of the 19th century, several historical barriers to international trade had been substantially reduced: transport costs fell with the spread of steamships and railways (O'Rourke and Williamson, 1999), communication frictions declined with the telegraph (Steinwender, 2018), and postal infrastructure was internationalized through the Universal Postal Union established by the Treaty of Bern in 1874. Shipping times were reduced significantly (e.g. by 79% between Liverpool and New York (Hatton, 2025)), which also induced reduction in freight rates by up to 30% between 1870-1914 (Estevadeordal et al., 2003). International trade was increasing throughout the second half of the 19th century until WWI, before plummeting throughout the world wars (Klasing and Milionis, 2014).

Building on these technological and infrastructural innovations, trade institutions emerged facilitated by domestic commercial chambers and ministries, resulting in increased foreign representation for national business interests (Rose, 2007). In parallel, businessmen created new organizations for information sharing for network building, including export associations and private networking events (Eichenberger and Péry-Kaulmann, 2022). In addition, trade fairs and museums of commerce helped businesses finding trustworthy buyers and suppliers.

Due to the increasing demand for information on foreign markets, collecting information on foreign markets became widespread across industrialized countries. One of the most important institutions for increasing trade between countries and collect market intelligence was consular offices. As opposed to embassies, one country had multiple consular offices, usually presented in important commercial ports and trading hubs (de Goey, 2014). By the turn of the twentieth century, several hundred such offices operated across the globe. Figure B.4 in the Appendix shows that although nearby markets had more offices than distant ones, remote destinations — including East Asian and South American countries — were also part of the consular network. Consuls collected information about tariffs, foreign market conditions, exchange rates and currencies, foreign trade associations, postal and telegraph fees, transportation costs including shipping rates and freight companies, and on domestic economic conditions in general. The Export Compass used in this project is an example of information collection used to foster trade relations.

According to Malfatti di Monte Tretto (1904), beyond these periodic reports, consuls were obligated to file ad hoc special reports on a defined set of topics including railway developments, public tenders and contract announcements, fraudulent firms, tariff changes, and bankruptcy notices. Crucially, consuls were required to respond to commercial inquiries about foreign buyers, with standardized instructions specifying the elements of a creditworthiness assessment, the choice of references, the required format, and the procedural distinction between Austrian and Hungarian enquirers reflecting the dual-monarchy's constitutional structure.

The institutional pipeline through which consular intelligence reached exporters is also specified in Malfatti di Monte Tretto (1904). Periodic trade reports were formally routed from consular offices through the Ministry of Foreign Affairs to the two trade ministries, and subsequently published

via the k.k. Handelsmuseum in Vienna, the Hungarian periodical *Consuli Közlemények*, and the *österreichisch-ungarische Konsularkorrespondenz* (Malfatti di Monte Tretto, 1904, p. 209). The training of future consular officials was itself organized around commercial competence: consular attachés were systematically assigned to Chambers of Commerce in major trading cities—including Vienna, Budapest, Trieste, Fiume, Prague, Brünn, and Reichenberg—where a designated chamber expert was responsible for their commercial and trade-policy education. This institutional architecture confirms that the Export Compass and the *Berichte der k.u.k. Konsularämter* were not ad hoc publications but downstream products of a mandated, hierarchically organized intelligence system, in which the identification and assessment of foreign buyers was a formal and regulated duty of the consular service.

The information flow specific for the buyer lists is shown in Figure 1. Consular offices collected information on local importers, which, through the institutional pipeline were synthesized in the Export Compass. Firms in Austria-Hungary had the opportunity to access and book and learn about the importers, and identify potential targets for their sales. In case the address and contact details were not indicated in the book (which often was the case), firm could reach out directly to the consular offices, as these foreign representatives had an obligation to supply information to firms upon request. Once firms received additional information directly from the consular offices, they could contact foreign importers though agents, telegraph, or post.

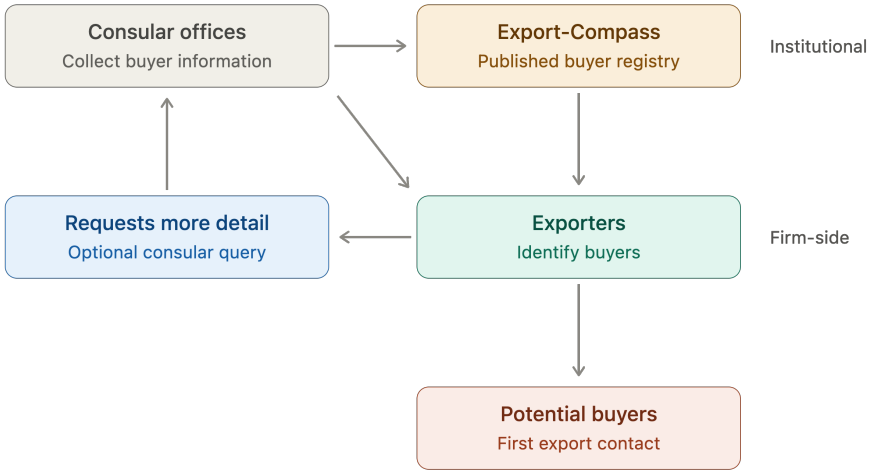


Figure 1: Information flow between firms and institutions

3 Constructing a historical firm-level panel

The empirical analysis builds on a newly constructed dataset from the *Austro–Hungarian Export-Compass* (*Export-Compass*), a government-adjacent trade publication produced by export-promotion institutions in cooperation with chambers of commerce and the consular network. An early version of the directory, limited to Viennese firms, appeared in 1888. By 1890, the publication expanded to cover firms throughout Austria–Hungary and organized them by industry and chamber jurisdiction. The 1890 volume also marks the first systematic inclusion of information on foreign importing firms (potential buyers).

The *Export-Compass* served two complementary purposes. First, it helped foreign buyers identify suppliers in Austria–Hungary by providing an organized directory of exporting firms and their product scope. Second, it supplied Austro–Hungarian firms with market intelligence intended to facilitate entry into, and continued operation in, foreign markets. Beyond firm directories, the publication compiled practical information for international commerce, including tariffs, exchange rates, measurement systems, consular locations, and contact information for export associations and museums of commerce.

A central component of the publication was its buyer-information service. Each year, different consular offices were instructed to collect or update lists of importing firms in a specific set of destinations, including the firms' location and the product groups they imported. Coverage rotated across world regions over time. This paper focuses on the volumes from 1890 to 1898: 1890 is the first year with comprehensive coverage of exporters across Austria–Hungary, and by 1895 the buyer lists span all major regions covered by the publication cycle. From 1896 onward, the reporting sequence restarted with the same region covered in 1890, consistent with a rotating coverage design.

3.1 Domestic exporting firms

Between 1890 and 1898, exporting firms are observed annually, yielding nine cross-sections that can be linked into a firm-level panel. The data contains 13,190 unique firms, of which 7,545 report at least one export destination at least once during the period. For each firm-year, the directory records the firm name and address, a description of exported products, and the destination countries served in that year. In some cases, entries also include additional details such as exhibition participation, advertised specialties, or secondary business locations. Since some chambers of commerce ceased reporting export destinations in 1892, only firms are kept from the 1890-91 sample that exist beyond 1892 to improve on the balance of the sample. For this reason, 5,500 firms remain in the final sample that is used for estimation. Including all firms from 1890-91, however, does not change the results significantly.

Table 1: Firm-Year Destination Coverage and Category Counts by Year

Year	Firms	Cov. rate	Avg. dest.	Dest. countries	Categories
1890	2,960	82.4%	3.0	58	413
1891	5,019	78.5%	2.8	59	469
1892	5,423	48.6%	2.6	59	488
1893	6,652	39.3%	2.5	59	518
1894	7,534	34.1%	2.4	55	533
1895	8,190	32.9%	2.4	55	534
1896	8,719	33.3%	2.4	57	545
1897	9,190	31.7%	2.5	57	544
1898	8,510	32.4%	2.4	56	542

Notes: *Firms* is the number of distinct firms listed in that edition; counts are of distinct firms and therefore exclude duplicate listing entries. *Cov. rate* is the share of those firms naming at least one export destination. *Avg. dest.* is the mean number of distinct destination countries named. *Dest. countries* is the number of distinct destination countries named by any firm that year. *Categories* is the number of distinct product categories assigned that year under the cleaned German category vocabulary. The decline in the coverage rate after 1891 reflects how completely destinations were recorded across editions rather than firms ceasing to export.

3.2 Foreign importing firms (potential buyers)

The buyer lists cover importing firms located abroad and are not restricted to importers of Austro–Hungarian goods. Instead, they document firms with ongoing import activity in the destination market. For each importer, the directory reports the firm's location and the products or product categories imported; in some cases, it also records the nationality of the owner. If an importing firm had an affiliation with Austria–Hungary—for example, through Austro–Hungarian protection or a formal connection to the monarchy—this was explicitly noted in the registry. In addition, the market activity of the importing firm is indicated: for example, *en detail* firms sold their products to consumers, while *en gross* firms purchased goods in large quantities and supplied smaller local firms. *Agents* and *commissioners* often acted as representatives or merchants of larger organizations.

3.3 Entity resolution across years

One challenge is to find firms across datasets, as there is no identification number available to match firms. To create a panel, the names of firms are used to match them across years. Since data collection was relatively consistent, many firms are successfully matched between years. The dataset covers between 3,173 and 10,320 firm-year observations per edition after removing within-year duplicates (see Table D.1).To track firms over time, names are matched across all year-pairs (e.g. 1890-91 or 1893-96) using exact and fuzzy string matching. Of the 151,708 total cross-year matches identified, 87.6% are exact and the remainder are fuzzy with a mean match score of 99.36—indicating that most name discrepancies reflect minor spelling variations rather than substantive mismatches. Match quality remains consistently high for consecutive year pairs, with mean scores

Table 2: Descriptive Statistics: Import Demand Signals by Year

	N		Share of firms (%)	
Year	Firms	Products	Merchants	Austro-Hung.
1890	8,586	157	6.5	6.5
1891	4,489	186	8.6	2.2
1892	1,827	125	20.2	4.5
1893	3,592	173	12.1	0.9
1894	3,793	208	14.4	2.8
1895	1,607	128	21.3	1.7
1896	10,116	249	6.9	4.0
1897	4,027	191	5.3	2.3
Total	38,037	339	9.3	3.7

Notes: more information on which countries were covered in these years are found in Table 4.

all exceeding 99.6 (see Table D.2). Firm persistence varies considerably across the panel. Nearly 30% of firms appear in only a single edition, pointing to high turnover in the exporter population, while 8.1% are observed in all nine editions and form a stable core of persistent exporters.

3.4 Treatment and outcomes

Treatment is defined using firms' observed export product mix in a given year. On the demand side, each destination's consular list of importing firms is coded into roughly 550 product categories. Exporting firms are then matched to destinations whenever at least one importer in that destination is recorded as demanding the firm's exported product(s). For instance, the Bulgarian lists contain 17 importing firms classified as buyers of candles in 1890; consequently, every Austro-Hungarian candle exporter in that year receives a "signal" indicating potential buyers in Bulgaria. Because firms may export multiple products, a single firm can receive multiple signals—either from the same destination across different products or from different destinations altogether.

This procedure imitates the information-gathering process available to contemporary firms: having purchased the Export Compass, a firm would read through the relevant consular lists and identify potential buyers. The LLM matching replicates this process by linking firms' exported products to importers' demanded products [3]. Table 3 reports the annual number of signals and their prevalence in the firm–year–destination panel (i.e., the share of firm–year–country cells with at least one signal). Signal counts vary over time since each years covered a different region and number of countries, and the density of consular network also varied across countries. Further information about data cleaning can be found in Appendix C.

[3]See an example in Appendix Figure B.3

Table 3: Yearly summary statistics at the firm–destination–year level

	Panel size		Share of observations (%)			Firms
Year	Obs.	Firms	Signal_t	Export_t	Export_{t+1}	signaled (%)
1890	63,920	1,598	10.4	7.4	6.9	87.6
1891	112,280	2,807	7.7	7.2	5.5	88.3
1892	111,480	2,787	6.0	6.8	4.9	80.4
1893	120,720	3,018	3.0	6.2	4.6	83.8
1894	124,200	3,105	6.8	5.8	4.6	86.2
1895	131,760	3,294	3.7	5.7	4.6	67.4
1896	140,160	3,504	10.4	6.0	4.7	88.6
1897	142,280	3,557	4.6	6.0	4.7	84.5
Total	946,800	—	6.3	6.3	4.9	83.0

Notes: The table is constructed at the firm–destination–year level. Obs. counts firm–destination–year observations and Firms counts unique firms observed in that year; the firm count is not additive across years, so no total is reported. The three share columns give the percentage of firm–destination–year observations with $\text{signal}_t = 1$, $\text{export}_t = 1$ and $\text{export}_{t+1} = 1$ respectively. Firms signalled gives the percentage of firms observed in that year with at least one signalled destination, and is therefore a share of firms rather than of observations. The Total row pools all 946,800 observations from 1890 to 1897; the corresponding figure for firms signalled pools firm–year observations. 1898 is observed for export status but lies outside the signal data window and is excluded.

4 Empirical Strategy

4.1 The quasi-random variation

The empirical strategy exploits quasi-experimental design in the availability of buyer information generated by the annual publication cycle of the *Export-Compass*. Through its global consular network, Austria–Hungary collected and published directories of importing firms (potential foreign buyers). Importantly, the registry did not cover all destinations every year: instead, information provision rotated across world regions in a staggered manner, such that in a given year only a subset of destinations was covered. This institutional feature creates year-to-year variation in which destination markets become "information-rich" for exporters. There are two main reasons to believe that listing buyers is exogenous to export potential: first, using regions, instead of countries, strongly indicates an administrative decision, as countries covered in each year are unlikely to have a correlated product-specific demand shock. Second, the rotating nature of the region coverages also suggest a simple methodological choice instead of careful consideration of economic dynamics or lobbying. As shown in Table 4, in 1896, the same countries are covered as in 1890, 1897 covers the countries from 1891, albeit missing Spain, Portugal, and Malta. 1898 covers the countries from 1892, except replacing Hongkong with Korea. Since there was no publication in 1899, and the publication of destination countries were ceased in 1900, 1898 is the last country covered in the analysis.

Table 4: Countries covered

Year	Countries covered
1890	Egypt, Bulgaria, Greece, East-Rumelia*, Serbia, Romania, Turkey
1891	Italy, Spain, Portugal, Algeria, Morocco, Tunisia, Malta*
1892	Indies, Hongkong*, China, Japan, Australia
1893	Canada, USA
1894	Switzerland, Russia, Sweden, Norway
1895	Chile, Peru*, Paraguay*, Argentina, Honduras*, Cuba*, Uruguay*, Brazil
1896	Egypt, Bulgaria, Greece, East-Rumelia*, Serbia, Romania, Turkey
1897	Italy, Morocco, Algeria, Tunisia
1898	Indies, China, Japan, Korea, Australia

Notes: Countries marked by * are covered in the Export Compass, however, due to the negligible export to these countries, they are not included in the analysis. Countries in 1898 are also not included, since there was no publication of Export Compass in 1899, the next and last version including export destination countries is in 1900.

I interpret the introduction of buyer listings for a destination country as an information shock that lowers search and matching frictions for market entry. Firms are differentially exposed to this shock because the information is product-specific: a firm is treated in destination d and year t if

the registry contains buyer information relevant to the firm's product scope for that destination. As explained in detail above, the main identifying variation therefore comes from changes over time in the set of covered destinations, interacted with firm-level product relevance.

4.2 Baseline Specification

I estimate how the provision of buyer information in year t affects the probability that firm f enters destination d in year $t+1$ using a panel at the firm–destination–year level:

$$\text{export}_{fd,t+1} = \beta\, \text{signal}_{fdt} + \gamma_{dt} + \alpha_{ft} + \phi_{fd} + \varepsilon_{fdt}, \tag{1}$$

where $\text{export}_{fd,t+1}$ is an indicator for export status for firm f at destination country d at time $t+1$. Signal_{fdt} indicates whether the *Export-Compass* lists buyers relevant to firm f's product scope in destination d in year t. Destination-year fixed effects γ_{dt} absorb time-varying demand and trade-cost shocks in each market. Firm-year fixed effects α_{ft} capture changes in firms' overall export capacity and supply conditions. Firm–destination fixed effects ϕ_{fd} account for persistent bilateral relationships such as geography, networks, or long-run market fit. In some specifications, I additionally control for lagged export status Export_{fdt} to account for persistence in exporting. Standard errors are clustered at the firm level, however, alternative clustering choices are shown with the main results.

Under the identifying assumption that the rotation of destination coverage is orthogonal to firm-specific export shocks conditional on the fixed effects, β recovers the causal effect of access to state-provided buyer information on subsequent export entry. This design addresses the two main confounds in observational trade data—systematic differences in firm export ability and heterogeneous destination-level shocks—by exploiting within-firm, within-destination variation in information availability over time. The fixed effects absorb 72.6% of the variation, thus the remaining is used to estimate the causal effect of the signal.

4.3 What the design absorbs: key confounding factors

The empirical specification is designed to address the major confounding concerns that arise in observational studies of exporting.

First, destination markets experience time-varying demand and trade-cost shocks that simultaneously affect export outcomes and the perceived usefulness of foreign market intelligence. Destination-year fixed effects γ_{dt} absorb any shock common to all firms exporting to a given destination in a given year: macroeconomic booms or recessions (e.g., rising demand in Egypt in 1892), exchange-rate movements, political events, tariff changes, new shipping connections, or diplomatic and trade-promotion initiatives. The estimated effect is therefore not driven by treated destinations becoming more attractive for reasons unrelated to information provision.

Second, firms differ in export capacity over time due to changes in productivity, ownership, technology adoption, or scale. Firm-year fixed effects α_{ft} control for such time-varying firm-level factors, so identification does not rely on comparing structurally different firms (e.g., fast-growing firms) to others, but on within-firm reallocation of export activity across destinations.

Third, exporting is shaped by persistent bilateral factors specific to a firm–destination pair: geographic distance, language, established commercial networks, diaspora links, and long-run product–market fit. Firm–destination fixed effects ϕ_{fd} absorb these time-invariant determinants, so the coefficient on buyer information is identified from changes within the same firm–destination pair over time rather than from cross-sectional differences across pairs.

Finally, export participation is highly persistent: firms that exported to a destination in the past are more likely to continue due to sunk costs, learning, or established relationships. Lagged export status export_{fdt} accounts for this path dependence and separates the effect of new information from mechanical continuation.

Taken together, the identifying variation is sharply delimited: the effect of buyer information is estimated by comparing firms whose products match the newly listed buyers to firms in the *same destination and year* without such a match, within the same firm–destination pair over time. Threats to identification must therefore operate at the product–destination or firm–destination level—narrower than in a standard destination-level design. The next subsection discusses these remaining threats.

4.4 Threats to identification and remaining concerns

Non-random coverage and selection of destinations. The primary concern is that coverage rotation was not exogenous: the compilers might have prioritized booming markets or destinations where Austro-Hungarian exports were already expanding. Given the design, selection of entire destinations cannot drive the estimates; the relevant threat is selection on *product-specific* prospects—for instance, if consular offices collected more buyers of glassware in Serbia precisely because glass manufacturers were preparing to enter that market. Two pieces of evidence speak against this. First, the rotation schedule was institutionally predetermined: the Compass followed a six-region cycle between 1890 and 1895, and the 1896 preface explicitly states the completion-then-restart rule governing re-coverage. Contemporary reviews described the rotating coverage as "eine Specialität des Export-Compass," a fixed editorial design rather than a response to current trade conditions. Second, if coverage tracked latent product-level demand growth, buyer counts from the *following* year's edition should predict entry today; Table A.2 shows they do not.

Reverse causality and lobbying. A related concern is that firms or trade associations requested buyer lists for particular destinations, so that coverage responded to firms' export plans. The institutional setting leaves little room for this channel. The buyer lists were compiled by k.u.k. consular offices abroad and released by the Ministry of Foreign Affairs to the publisher; individual

exporters had no role in selecting destinations, and the rotation schedule was set in advance for the full cycle. The publication timing reinforces this: entries were current to the end of July and published in October, so the information could affect export behavior only from the following year onward—which is why the outcome is measured at $t+1$—leaving no scope for coverage to respond to contemporaneous entry decisions.

Anticipation. Because the rotation schedule was announced, firms could in principle anticipate coverage of a destination and delay entry until the buyer lists appeared, or prepare entry in advance. Anticipatory delay would bias the estimates upward; advance preparation would generate effects before publication and bias them downward. The lead placebo in Table A.2 bounds both concerns: export behavior shows no response to information that has not yet been published.

Bundling with other trade-promotion measures. Destination-wide policy initiatives coinciding with publication are immaterial by the same logic as above; the residual concern is product- or firm-targeted intervention bundled with the Compass. Here the nature of the treatment matters: the Compass was an informational tool published privately by Alexander Dorn's Volkswirtschaftlicher Verlag, providing buyer identities, locations, product scopes, and commercial functions—no financial support, no logistical assistance, and no firm-specific services accompanied publication. Any effect of the listings therefore reflects information, not a bundled program.

Measurement error in treatment assignment. Assigning treatment requires mapping firm product scopes to listed buyers' product descriptions, which is noisy: product categories in the registry are sometimes broad, and descriptions incomplete. Misclassification of this kind attenuates the estimates toward zero, so the reported effects are rather conservative.

Spillovers and information diffusion. The published registry was a public good: firms could share the lists, and intermediaries could act on them. Such diffusion may contaminate the control group—untreated firms partially receive the treatment—and biases the estimates toward zero. In this case, the coefficients should therefore be read as a lower bound on the full effect of information provision.

Sample selection and outcome measurement. The outcome is constructed from the Compass's own exporter directory, so the specific concern is whether the quality of *exporter-side* reporting co-moves with *buyer-side* coverage rotation—if destination coverage years coincided with more complete recording of export destinations, the estimates would partly reflect measurement. The exporter directory was compiled annually for the full set of firms, on a cycle independent of the buyer-side rotation.

5 Results

5.1 Main results

Table 5 shows the effect of receiving a signal at time t on export status at time $t+1$. Standard errors are clustered at the firm level; the columns add fixed effects progressively. Without any fixed effects (column 1), the signal coefficient is 2.89 percentage points. Adding country–year fixed effects (column 2) reduces it to 1.31, adding firm–country fixed effects (column 3) to 0.301, and the saturated specification with all three pairwise fixed effects (column 4) yields 0.289 percentage points, significant at the 1% level.[4] The decline across columns is expected: most of the raw correlation between signals and subsequent exporting reflects destination-year conditions and persistent firm–destination relationships, which the fixed effects progressively absorb. The sharp drop between columns (2) and (3) is informative in itself: once firm–destination pairs are held fixed, three quarters of the remaining correlation disappears, indicating that the bulk of the naive association is cross-sectional sorting of firms into markets they were already serving rather than time-varying destination conditions. The saturated estimate—identified solely from comparing firms with and without product-matched buyers within the same destination-year, within the same firm–destination pair over time—is the preferred specification. Relative to the baseline entry probability of 4.9 percent, it implies that receiving a signal raises the probability of exporting to the destination in the following year by roughly 5.9 percent.

Table 5: Effect of consular buyer information on next-year exporting

	(1)	(2)	(3)	(4)
	export$_{t+1}$			
signal$_t$	2.84***	1.19***	0.275**	0.287***
	(0.135)	(0.197)	(0.113)	(0.104)
	[0.105]	[0.150]	[0.0879]	[0.0842]
	{1.50}	{0.339}	{0.163}	{0.130}
Observations	946,800	946,800	934,240	934,240
R^2	0.001	0.113	0.795	0.819
Firm–country FE	No	No	Yes	Yes
Firm–year FE	No	No	No	Yes
Country–year FE	No	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t+1$, thus units are percentage points. Standard errors clustered at the firm level in parentheses; two-way clustered at the firm and destination–year level in square brackets; clustered at the destination level in braces, with wild-cluster-bootstrap p-values. Clusters: 7,545 firms, 320 destination–years, 40 destinations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

[4]The fixed effects absorb 72.6% of the variation in signal$_t$; the coefficient is identified from the remaining 27.4%, which corresponds to variation in signaling across destinations within a firm–year and across years within a firm–destination pair.

Standard errors are clustered at the firm level throughout. A firm chooses its export markets jointly rather than destination by destination, and unobserved firm characteristics—e.g. managerial capacity, credit access, or an established shipping agent—persist across the sample and affect every destination the firm faces, thus errors are correlated across all cells belonging to the same firm. Clustering on the firm accommodates this dependence in both directions while leaving 7,545 independent units, comfortably above the range in which cluster-robust variance estimators become unreliable. The correction matters: heteroskedasticity-robust standard errors that ignore within-firm dependence are roughly half as large, and would overstate the precision of the estimate by a factor of about two. Because the product composition of each edition's buyer list is common to all firms facing a given destination in a given year, the last two rows of each column also report standard errors that are two-way clustered at the firm and destination–year level and clustered at the destination level; the latter is a conservative bound with only 40 clusters, so its p-value comes from a wild cluster bootstrap. The estimate remains significant under all clutering specifications.

5.2 Placebo treatment

Table 6 tests whether the baseline effect of signal_t on export_{t+1} holds when using a placebo signal for receiving a counterfactual buyer information before the actual information arrives. Since both the list of exporters and importers come from the same source, it should not be possible for an exporter to list its destinations before learning about potential importers. Even if an exporter updates its export destination list, the updated version should appear in the Export Compass in the subsequent year.

The leads show no anticipatory effects — signal_t on current export and signal_{t+1} on future export, and future signal on past export are uniformly insignificant statistically. The small magnitude of the coefficient also suggest insignificant economic effect, not only a result from decreased sample size. These are clean falsification results supporting the importance of signals on export status a year later.

5.3 No effect for signals received two years or before

To analyze whether signals have longer-term effect, I regress the signals received one or two years before the baseline signal. For this reason, singleton firm–country cells (17,400 observations) are dropped. In Table 7, the sample is restricted to outcome years for which the relevant lag window is observed: 677,120 observations where signal_{t-1} is available and 545,240 where signal_{t-2} is also available. Columns (1) and (4) re-estimate the baseline specification on these restricted samples without lags, so that the effect of adding lagged signals is separated from the effect of the sample restriction itself. Lagged signals are set to zero for outcome years whose lag window precedes the start of the signal data, on the grounds that no signaling opportunity existed before 1890.

The effect of receiving relevant buyer signal is concentrated at a one-year horizon. In column

Table 6: Effect of buyer-information placebo signals on next-period exporting

	(1)	(2)	(3)	(4)
		export$_{fd,t+1}$		
signal$_{fdt}$	0.248***	0.228**	0.289***	0.235*
	(0.0896)	(0.0951)	(0.110)	(0.127)
signal$_{fd,t+1}$		−0.0980		−0.152
		(0.0970)		(0.127)
signal$_{fd,t+2}$				−0.155
				(0.135)
Mean of dep. var.		0.49		
Observations	723,520	723,520	510,400	510,400
R^2	0.912	0.912	0.909	0.909
Sample	t+1	t+1	t+2	t+2
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(2), the estimate is 0.427 percentage points, indicating that the firm–destination pairs observed over longer stretches of the panel respond more strongly to signals; the comparison of columns (1) and the baseline in Table 5 isolates this composition effect from any role of the lag structure. Within the restricted samples, adding lagged signals leaves the contemporaneous coefficient essentially unchanged—0.427 to 0.412 percentage points in column (2) and 0.442 to 0.439 in columns (4)–(5)—while the lagged terms themselves are small and statistically indistinguishable from zero: −0.0696 percentage points for signal$_{t-1}$ in column (2), and 0.078 and 0.125 percentage points for signal$_{t-1}$ and signal$_{t-2}$ in column (4), against standard errors of 0.134 to 0.163. Taken together, the estimates indicate that a signal affects export entry in the year immediately following its receipt, with no detectable additional effect from signals received two or three years earlier. Columns (3) and (6) estimates the persistence in a different way: while in columns (2) and (5) signals appear only the years they buyers lists are published and disappear the following years, column (3) assumes that the buyer lists are still available since firms could access the previous-year version of the Export Compass. In this case, a firm-country-year cell is treated if a signal is received in year t or $t-1$. The results are consistent with the findings in column (2): if signals are assumed to stay on for another year, the estimates become noisier, and the magnitude also drops to 0.183 percentage points. Similar results are found in column (6), where signals are assumed to stay on for 3 years.

Table 7: Persistence of signal effect beyond 1 year

	(1)	(2)	(3)	(4)	(5)	(6)
			export$_{fd,t+1}$			
signal$_{fdt}$	0.427***	0.412***		0.442***	0.439***	
	(0.118)	(0.127)		(0.131)	(0.147)	
signal$_{fd,t-1}$		-0.0696			-0.0478	
		(0.133)			(0.162)	
signal$_{fd,t-2}$					0.0588	
					(0.138)	
signal$^{(2)}_{fdt}$			0.183*			
			(0.109)			
signal$^{(3)}_{fdt}$						0.0695
						(0.122)
Mean of dep. var.			.049			
Observations	677,120	677,120	677,120	545,240	545,240	634,560
R^2	0.849	0.849	0.849	0.858	0.858	0.835
Sample	$t-1$	$t-1$	$t-1$	$t-2$	$t-2$	$t-2$
Firm–country FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.4 Past export status and export market entry

Table 8 reports specifications that interact and restrict the sample by past export status to the specific destination country in the year the signal is received. Column (1) augments the main specification from the last column of Table 5 with an indicator for exporting to the destination in year t: the signal coefficient is essentially unchanged, neither statistically nor in magnitude. Including a lagged dependent variable in a fixed-effects panel is normally discouraged because of the well-known Nickell bias, but the concern is muted here, since export status in t is predetermined with respect to the signal by the construction of the Export Compass.[5] Column (2) interacts the signal with past export status. The signal main effect becomes insignificant and falls to roughly one eighth of its column (1) value, while the interaction term is large and precisely estimated: firms already serving the destination respond by about 4.2 percentage points, whereas firms not yet serving it show no measurable response. Since incumbent firm–destination pairs are a small minority of the sample, the pooled estimate of 0.30 in column (1) is essentially this large effect diluted by a mass of non-responding observations.

Columns (3) and (4) restrict the sample to observations with no past export status to the destination, so that the estimates capture export market *entry* rather than the intensive margin. This restriction retains 93.6% of the full sample, confirming that only a small share of firm–destination–

[5]Buyer information and contemporaneous export status are published in the same volume, so any change in export behaviour induced by the Export Compass should first appear in the following edition, unless a firm obtained the buyer information independently and at least a few months before the relevant volume was published.

Table 8: Past export status and export market entry

	(1)	(2)	(3)	(4)
			$\text{export}_{fd,t+1}$	
signal_{fdt}	0.297***	0.0385	0.0306	
	(0.100)	(0.0882)	(0.0368)	
export_{fdt}	29.0***	28.6***		
	(1.23)	(1.23)		
$\text{signal}_{fdt} \times \text{export}_{fdt}$		4.19***		
		(0.746)		
signal_{fdt} (thick)				0.0944**
				(0.0442)
Observations	934,240	934,240	874,098	874,098
R^2	0.829	0.829	0.463	0.463
Sample	Full	Full	$\text{export}_t = 0$	$\text{export}_t = 0$
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t+1$, so units are percentage points. signal_{fdt}(thick) is defined as signals that list at least 5 buyers. Columns (3) and (4) restrict the sample to observations where there is no current export. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

year observations already involve exporting to that destination. Column (3) mirrors column (2) in finding no measurable effect of a signal on entry. Column (4) replaces the signal indicator with a "thick" signal that equals one only if at least five buyers are listed, thereby discarding thin signals that name only one or two buyers. The entry effect now becomes significant at the 5% level, with a small but measurable magnitude of roughly one third of the main effect in Table 5. This pattern suggests that firms need to be presented with several potential buyers before entering a new market, consistent with higher fixed costs of entry into untapped destinations.

5.5 Other robustness checks

5.5.1 Firms with export destination data for at least three years

Appendix Table A.1 re-estimates the baseline on a sample restricted to firms reporting destination-level information in at least three years. This restriction addresses potential selection from the contraction of destination coverage after 1891: if firms with incomplete destination reporting differ systematically from consistently reported firms, the baseline estimates could partly reflect reporting rather than behavior. The results are close to the baseline throughout: the signal coefficient is 3.03 percentage points without fixed effects (column 1), 1.34 with country–year fixed effects (column 2), 0.303 once firm–country fixed effects are added (column 3), and 0.248 in the saturated specification (column 4), compared to 2.89, 1.31, 0.301, and 0.289 in the full sample. The restricted-sample estimates are, if anything, marginally larger in the first three columns; only the saturated

estimate falls, by roughly a seventh, and it remains significant at the 5% level. Some attenuation there is unsurprising given that the restriction removes roughly a sixth of the observations and, by construction, the short-lived firms whose entries and exits contribute much of the identifying variation. The stability of the point estimates indicates that selective reporting does not drive the baseline results.

5.5.2 Continuous treatment

Appendix Table A.2 exploits the intensive margin of the information treatment: the number of product-matched buyers listed for a destination, rather than a binary coverage indicator. Column (1) shows that one hundred additional listed buyers raise the probability of exporting to that destination in the following year by 0.66 percentage points—about 0.0066 percentage points per buyer, so that a firm facing ten additional listed buyers is roughly 0.07 percentage points more likely to enter—conditional on current export status and the full set of pairwise fixed effects. The effect is precisely estimated despite the demanding specification, which absorbs all destination-year demand shocks through country–year fixed effects: identification comes from comparing firms with more versus fewer product-matched buyers within the same destination-year cell. Column (2) shows that the relationship is not an artefact of the largest listings, with a doubling of the number of matched buyers raising the entry probability by roughly 0.10 percentage points, and column (3) confirms that what matters is the total volume of matched buyers rather than the size of the single largest product signal: conditioning on the maximum count leaves the coefficient on the total intact, while the maximum itself enters negatively and insignificantly. Column (4) reports the placebo. If the estimated coefficient reflected unobserved firm–destination demand trends—firms expanding toward destinations where buyer demand was growing—then buyer counts published in the following year's *Export Compass*, which firms could not yet observe, should also predict entry. They do not: the coefficient on the one-year lead is -0.145 with a standard error of 0.182, statistically indistinguishable from zero and estimated precisely enough to exclude an effect anywhere near the contemporaneous one. Export behavior responds to buyer information only once it is published, consistent with the information channel and inconsistent with firms anticipating future demand conditions. The smaller sample in column (4) reflects the loss of the final panel year required to construct the lead.

6 Mechanisms

6.1 Product-specific market demand vs. information

To understand how information on potential buyers affects export market entry and participation, it is important to account for the possibility of product-specific demand booms in the markets from which signals are received. Instead of using the information about potential buyers, Austro-

Hungarian firms may interpret the number of buyers as an indicator of increased demand for specific products instead of acting on the information received. In addition, the probability that a signal appears in it may still be correlated with a positive demand shock for the goods that signal concerns.

Table 9: Market reports and their interaction with buyer signals

	(1)	(2)
	export_{t+1}	export_{t+1}
signal_t	0.277***	0.274***
	(0.104)	(0.104)
$\text{signal}_t \times \text{AH dominant}_t$	3.85**	
	(1.56)	
$\text{signal}_t \times \text{encouraging}_t$		1.03**
		(0.448)
Observations	934,240	934,240
R^2	0.819	0.819
Firm–country FE	Yes	Yes
Firm–year FE	Yes	Yes
Country–year FE	Yes	Yes

Notes: *Notes:* Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. Column (1) adds an indicator for the report describing Austro-Hungarian goods as dominant or established and column (2) for an encouraging tone. These indicators are nested within $signal_t$, so their coefficients are differentials relative to other reported categories and should not be added to it. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To address this, I control directly for product-specific market conditions using a second component of the *Export-Compass*: the *Specielle Artikel* sections, in which consular posts reported on individual commodities. I extract 426 such entries from the 1890–1894 volumes and their post-1894 continuations, covering 79 consular posts and 237 distinct product designations, and code each for whether the consul is urging Austro-Hungarian producers to enter or expand in that market. Detection is grammatical rather than sentiment-based: consular recommendations are written in Konjunktiv II (*wäre, könnte, dürfte, sollte*), the mood of the counterfactual recommendation, and this marker separates advice from mere description in a way a dictionary of positive terms does not. A short example, from Ancona in 1890, reads: "The popular Viennese pendulum clocks and the so-called Karlstein clocks could easily find a market here too, if they were cheaper and better adapted to local taste." Others additionally characterize competitive conditions and the standing of Austro-Hungarian firms in the local market, as in this report from Tripoli in 1892: "Domestic industry has only a small share in the import of silk goods; this could certainly be raised considerably if the manufacturers concerned tended this market rather better — which would be all the easier since Austria-Hungary produces extremely fine and reasonably priced oriental silks, which

could find a large market here." Matching these product designations to the buyer-list categories yields a signal for 65 of the 3,141 country–year–category cells; the coding agrees with hand coding in 87% of a validation sample, and the resulting measurement error attenuates the estimates below toward zero.

Table 9 splits the signal by the content of the contemporaneous consular report on the same product market. Column (1) adds an indicator for the consul describing Austro-Hungarian goods as already dominant or established in that commodity; column (2) an indicator for the report being encouraging, i.e. recommending entry or expansion. Both indicators are defined only where a report exists, so the coefficient on signal_t is the effect for product markets that were reported on but lacked the feature in question, and the second row is the additional effect of having it. Firms respond more strongly to signals from markets where Austro-Hungarian producers were already established (a further 3.85 percentage points, against a baseline of 0.28) and to signals accompanied by an encouraging report (a further 1.03 points). Neither drives the headline estimate: signals from product markets with no established Austro-Hungarian presence, and signals unaccompanied by an encouraging report, remain significant at 0.28 and 0.27 percentage points respectively, close to the saturated specification in Table 5. The differentials should be read as descriptive rather than precisely estimated: the position measure agrees with hand coding in only 47% of a validation sample, and both rest on the small subset of firm–destination pairs for which a matched product report exists.

6.2 Signal strength

In this section I differentiate between strong and weak signals using the mean and the median number of relevant potential buyers from a certain destination. Table 10 splits signaled observations into two bins at a cutoff in this measure; the omitted category in both columns is receiving no signal at all. Column (1) cuts at the median of the intensity measure among signaled observations in the estimation sample (10), which yields balanced cells. Column (2) cuts at its mean (25). Because the distribution is right-skewed, the mean cutoff defines a smaller and more sharply differentiated high-intensity group, at the cost of precision.

The ex ante expectation is that denser listings convey more information and should make finding a buyer easier, since more potential counterparties can be approached at low marginal cost. The main countervailing force is the novelty of information: destinations with many listed importers may be markets where demand for these products was already well known, or where consular offices had been active for years, so the marginal value of the listing may be lower.

Both columns show the same ordering, but they differ in how sharply it is estimated. Under the median cutoff, weak signals raise the probability of exporting to the destination in the following year by 0.22 percentage points and strong signals by 0.40 percentage points, against a baseline export rate of 4.9 percent; the difference between the two is not statistically significant ($p = 0.243$). Under the mean cutoff the weak-signal coefficient is essentially unchanged at 0.21 percentage points while

Table 10: Signal strength

	(1)	(2)
	$\text{export}_{fd,t+1}$	
	Median cut	Mean cut
$\text{signal}_{fdt}^{\text{weak}}$	0.222**	0.208**
	(0.104)	(0.099)
$\text{signal}_{fdt}^{\text{strong}}$	0.395**	0.666***
	(0.157)	(0.211)
$p(\text{weak} = \text{strong})$	0.243	0.015
Mean of dep. var.	.049	
Observations	934,240	934,240
R^2	0.819	0.819
Firm–country FE	Yes	Yes
Firm–year FE	Yes	Yes
Country–year FE	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t + 1$, so units are percentage points. Signal intensity is measured as listed buyers per matched product category. Signaled observations are split at the median (column 1) and the mean (column 2) of this measure among signaled observations in the estimation sample; the omitted category is no signal. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the strong-signal coefficient rises to 0.67, and equality is rejected ($p = 0.015$). The two columns are not independent tests of the same hypothesis but two cuts of the same data, and the pattern across them is informative: narrowing the high-intensity group raises its coefficient while leaving the weak-signal estimate untouched, which is what one would expect if the response continues to increase within the upper tail rather than stepping up at a single point.

Two conclusions follow, with different degrees of confidence. First, and robustly, even weak signals matter: identifying a small number of potential buyers per product line raises entry relative to no information at all, by an amount that is stable across both definitions of the cutoff. This speaks against a pure salience interpretation and is consistent with the low per-buyer cost of approaching a listed potential buyer. Second, and more tentatively, the response appears to scale with the density of information rather than merely its presence. The median split cannot reject a constant effect across intensity levels, so the evidence for a gradient rests on the narrower high-intensity group and should be read as suggestive. What the data do rule out is the opposite concern, that abundant listings signal already-saturated markets: the point estimates rise with intensity under both definitions, and nowhere is the high-intensity coefficient smaller than the low-intensity one.

6.3 Information Effects Across Destination Markets

Table 11 reports heterogeneity in the effect of buyer signals across destination-country characteristics, focusing on geography. Destinations are grouped into four categories: whether a country is

overseas and therefore reached across an ocean, whether it shares a border with Austria-Hungary (*neighbour*), whether it lies on the Balkan peninsula with established trade links (*balkan*), and whether it is located around the Mediterranean and readily accessible from the ports of Trieste and Fiume (*mediterranean*). These categories proxy for transport costs, distance, and the practical ease of serving a market.

The results suggest that the usefulness of buyer information depended on how actionable it was. In column (1), the baseline signal effect for non-overseas destinations is 0.36 percentage points and statistically significant, while the interaction with the overseas indicator is negative and of a similar order of magnitude (−0.271). Although the interaction is not statistically significant, the point estimates imply that the export response to signals from overseas markets was close to zero. This is consistent with firms being less able to act on information from distant destinations, likely because high fixed costs, transport frictions, and limited prior commercial presence made entry or reallocation more difficult, especially measure within a year. Furthermore, expected costs for importers might have prevented them entering business with distant and less-known exporters.

Table 11: Heterogeneity by destination country characteristics

	(1)	(2)	(3)	(4)
			$\text{export}_{fd,t+1}$	
signal_{fdt}	0.360**	0.195**	0.270***	0.112
	(0.159)	(0.0844)	(0.0921)	(0.135)
$\text{signal}_{fdt} \times \text{overseas}_d$	−0.271			
	(0.186)			
$\text{signal}_{fdt} \times \text{neighbour}_d$		0.292		
		(0.302)		
$\text{signal}_{fdt} \times \text{balkan}_d$			−0.0196	
			(0.353)	
$\text{signal}_{fdt} \times \text{mediterranean}_d$				0.446**
				(0.210)
Observations	934,240	934,240	934,240	934,240
R^2	0.828	0.828	0.828	0.828
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. The interaction categories are defined as follows: *overseas* denotes countries on other continents reached across an ocean; *neighbour* denotes countries bordering Austria-Hungary; *balkan* denotes countries with established trade links on the Balkan peninsula; and *mediterranean* denotes countries around the Mediterranean Sea readily accessible from Trieste or Fiume. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The interaction for neighboring countries in column (2) points in the opposite direction. The

coefficient is positive and economically sizeable—the implied total effect, at roughly 0.292, is more than double the effect elsewhere—but it is imprecisely estimated and cannot be distinguished from zero. The direction is nevertheless plausible: neighboring destinations were easier to serve because of shorter transport distances, lower transaction costs, and denser pre-existing commercial links, thus buyer information in such markets was more likely to translate into actual export entry or reallocation.

For Balkan destinations, column (3) shows an interaction that is rather negative. Signals from these markets were less effective than signals from elsewhere. This is rather surprising, since Balkan countries were important destinations for Austro-Hungarian exports. Even thought information from these countries is actionable, likely its novelty is less significant since the Austro-Hungarian Empire invested in uncovering business potential for the Balkan countries even before the lists for potential buyers were published in the Export Compass.

The clearest differential effect appears for Mediterranean destinations in column (4). The interaction is positive and statistically significant at the 5% level (0.446), while the baseline effect for non-Mediterranean markets is smaller and insignificant. The implied total effect for Mediterranean destinations, roughly 0.558, is about twice the baseline estimate reported in Table 5. This pattern is plausible given that these markets were connected to the monarchy through established maritime routes from Trieste and Fiume, which kept the cost of acting on buyer information low; at the same time, they may have been less saturated with existing commercial ties than immediate neighbors, leaving more scope for new matches.

Overall, Table 11 shows that the effect of buyer signals was concentrated in destinations where information could be turned into action at relatively low cost—most clearly the Mediterranean markets served through the monarchy's own ports, and, in terms of point estimates, neighboring countries—while it was muted for distant overseas destinations. The pattern therefore supports an interpretation of export promotion as reducing search and matching frictions conditional on market accessibility, rather than as a uniform information shock across all destinations.

6.4 The role of buyer types: intermediaries (merchants) and nationality

Table 12 shows the results from the heterogeneity analysis concerning buyer characteristics. The merchant share of a signal is calculated as $N_{\text{merchant}}/N_{\text{all}}$, i.e. the ratio of merchants to all buyers. As indicated above, those buyers are considered merchants that either purchase en gross (in large quantities) or directly indicate trader status (e.g. "Händler" or "Kommissionär"). Similarly, the Austro-Hungarian share is calculated as $N_{\text{Austro-Hungarian}}/N_{\text{all}}$, i.e. the ratio of Austro-Hungarian buyers to all buyers. The dummies *Merchant share high* and *Austro-Hungarian share high* are defined in two ways: for shares above 0.5 (columns 1 and 3) and for shares above the sample median (columns 2 and 4). Whether a buyer is Austro-Hungarian is explicitly indicated in the list

of buyers, and their categorization therefore required no further assumptions. Ex ante, a higher share of intermediary buyers should increase the probability of exporting to a given destination, since intermediaries purchase goods in larger quantities and are likely to have better information about further downstream demand. The role of Austro-Hungarian buyers is more ambiguous: on the one hand, exporters were more likely to already know these firms, so listing them may convey less novel information. On the other hand, establishing new commercial connections is easier with firms that operate under a shared legal framework, and communication should be smoother due to a common language, culture, and customs.

Table 12: The role of intermediaries and common citizenship for buyers

	(1)	(2)	(3)	(4)
			$\text{export}_{fd,t+1}$	
signal_{fdt}	0.255**	0.213*	0.291***	0.318***
	(0.106)	(0.111)	(0.106)	(0.0997)
$\text{signal}_{fdt} \times \text{high wholesaler share}_{fdt}$	0.643***			
	(0.231)			
$\text{signal}_{fdt} \times \text{high wholesaler share (med.)}_{fdt}$		0.253*		
		(0.143)		
$\text{signal}_{fdt} \times \text{high Austro-Hung. share}_{fdt}$			−0.153	
			(0.781)	
$\text{signal}_{fdt} \times \text{high Austro-Hung. share (med.)}_{fdt}$				−0.119
				(0.176)
Observations	934,240	934,240	934,240	934,240
R^2	0.818	0.818	0.818	0.818
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t+1$, thus units are percentage points. Columns (1) and (3) define high-share as more than 50% of buyers within a signal, while columns (2) and (4) use the sample median as the threshold. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The results confirm the ex ante expectation about merchant share. In column (1), when the merchant-share dummy equals zero, the signal effect is 0.255 percentage point increase, in line with the baseline findings. When the merchant-share dummy equals one, the total effect rises by approximately 0.643—a roughly threefold increase—indicating that signals dominated by wholesale intermediaries are substantially more informative for export entry decisions. The pattern is robust to the choice of threshold: with the median cut in column (2), the interaction remains positive and significant at the 10% level, and the implied total effect (about 0.466 percentage points increase) is more than double the effect of low-merchant-share signals, which remains significant on the 10% level. These results support the interpretation that firms buying in large quantities, as well as firms

specializing in international procurement, meaningfully facilitating export market presence. They likely maintain closer ties with local consular offices and may actively seek out potential foreign suppliers.

The evidence on Austro-Hungarian buyers is far less conclusive. In column (3), the interaction coefficient is negative and economically meaningful, however, it is imprecisely estimated and statistically insignificant, partially due to the very low number of signals containing high Austro-Hungarian share. Under the median threshold in column (4), the coefficient of the interaction terms is similar to the results in column (3). The two definitions thus give no consistent indication that signals dominated by Austro-Hungarian buyers were differentially effective. A natural reading is that the two ex ante forces offset each other: any advantage from contract enforceability and cultural–linguistic familiarity may have been neutralized by the lower novelty of information about compatriot firms that exporters were more likely to know already. Given the imprecision of the estimates, however, this interpretation should be treated with caution.

Table A.3 replaces the discretized dummies with the underlying continuous shares, which enter the specification directly rather than interactively. The coefficients are therefore not marginal effects at a threshold but the change in the export-entry probability associated with moving from a signal in which no buyer is of the relevant type to one in which every buyer is. Columns (1) and (2) introduce the two shares one at a time, column (3) includes both jointly, and column (4) substitutes the level of demand with its logarithm; all specifications retain the full firm $\times$ country, firm $\times$ year, and country $\times$ year fixed effects.

The continuous measure corroborates the dummy results for intermediaries. In column (3), shifting the buyer composition of a signal from entirely non-merchant to entirely merchant raises the probability of exporting to the destination in $t+1$ by 0.700 percentage points, significant at the 1% level, and the estimate is essentially unchanged when the Austro-Hungarian share is omitted (0.693 in column (1)) or when demand enters in logs (0.585 in column (4)). Relative to the baseline signal effect of roughly 0.255 percentage points, this is an increase of comparable magnitude to the threefold effect implied by the dummy interaction in Table 12, which is reassuring given that the two specifications exploit different parts of the variation: the dummies contrast coarse groups, whereas the continuous measure uses the full distribution of the share and is thus the more demanding test. The demand terms themselves remain positive and precisely estimated—0.00607 percentage points per additional unit of demand in column (1) and 0.127 percentage points per log point in column (4)—so the merchant-share coefficient is not simply absorbing the size of the signal or the level of destination demand, which would otherwise be the most obvious confound, since larger signals mechanically contain more wholesale buyers.

The Austro-Hungarian share again yields no usable evidence. The point estimate is negative throughout, ranging from -0.183 percentage points in column (2) to -0.525 in column (4), but the standard errors are of the same order as the coefficients themselves and none of the estimates approaches conventional significance. The pattern is thus consistent with the dummy specifications:

the sign is stable but the magnitude is not identified, and the confidence intervals comfortably contain both economically substantial negative effects and zero. Two features of the data caution against reading the negative sign as informative. First, signals with a high Austro-Hungarian share are rare, so the upper range of the distribution is thinly populated. Second, shares constructed from a small number of buyers are measured with considerable noise, and the resulting attenuation biases the coefficient towards zero—a concern that applies to both shares but bites hardest where the underlying counts are smallest. The joint reading of the two tables is therefore that the intermediary channel is robust to how buyer composition is defined, whereas the nationality channel remains an open question that the present data cannot settle.

6.5 Stronger persistence with merchant connections

Table 13 shows the results for the difference between signals with low vs. high merchant shares, as defined in Section 6.4. Column (1) shows the baseline interaction that was already presented in Table 12. The differential is estimated precisely enough to be significant at the 1% level, and it is the interaction rather than the un-interacted term that carries most of the average effect documented in Table 5.

Columns (2) and (3) add lagged signals, interacted with the merchant-share indicator measured at the same date. The non-interacted lags are small and statistically indistinguishable from zero (0.02 and -0.03 percentage points at $t-1$; 0.18 at $t-2$), consistent with the results in Table 7. The interaction terms, by contrast, remain positive at every horizon—0.39 percentage points at $t-1$ and 0.81 at $t-2$ in column (3)—so that the total effect for high-merchant-share destinations is 0.59, 0.36 and 0.98 percentage points for signals received at t, $t-1$ and $t-2$ respectively. Two features of the table caution against a stronger reading. First, the samples differ across columns (934,240, 604,200 and 415,080 observations) because each additional lag requires an additional year of observed signal history, so the attenuation of the contemporaneous interaction from 0.64 to 0.34 cannot be separated from the change in sample composition. Second, the confidence intervals on the interactions are wide relative to the differences between them, reflecting the low prevalence of the moderator. Overall, signals with high share of merchants specializing in wholesale seem to result in stronger commercial connections that persist well beyond one year. The difference between signals with low vs. high merchant shares are clear even if the sample is reduced to half of its original size.

6.6 Firm-size proxies

The next steps uncovering the mechanisms behind the utility of information is looking at proxies that indicate firm-size or at least firm capabilities. Three measures are used: the first distinguishes multi-plant firms, i.e. firms that have factories or offices in multiple locations. The second looks at firms that list multiple export destinations, indicating more experience in export markets. The third measure highlights firms with multiple product, likely proxying larger establishments that

Table 13: Persistence of effect for high merchant signals

	(1)	(2)	(3)
		$\text{export}_{fd,t+1}$	
signal_{fdt}	0.253**	0.254**	0.235*
	(0.105)	(0.110)	(0.138)
$\text{signal}_{fd,t} \times \text{high wholesaler share}_{fdt}$	0.604***	0.364*	0.379
	(0.224)	(0.213)	(0.262)
$\text{signal}_{fd,t-1}$		−0.0827	−0.163
		(0.133)	(0.135)
$\text{signal}_{fd,t-1} \times \text{high wholesaler share}_{fd,t-1}$		0.483**	0.409*
		(0.242)	(0.222)
$\text{signal}_{fd,t-2}$			0.109
			(0.128)
$\text{signal}_{fd,t-2} \times \text{high wholesaler share}_{fd,t-2}$			0.734***
			(0.278)
Mean of dep. var.		.049	
Observations	934,240	604,200	415,080
R^2	0.819	0.864	0.899
Firm–country FE	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t + 1$, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

have a wider portfolio of goods exported.

It is unclear ex ante whether buyer information is more or less useful for these firms. On the one hand, larger or more experience firms can act on the information easier, having established networks with agencies and higher capabilities to ship goods. On the other hand, for the same reasons, these firms may already know about all potential buyers, since their channels should be more exhaustive and leave little room for information discovery. Note that every firm in the sample has the capability to export, as they Export Compass only collected information on firms that exported at least to one country.

Table 14 shows the result for the three proxies; the level terms are absorbed by the firm–year fixed effects, so only the interactions are identified. All three interactions are positive: the signal effect is 1.11 percentage points larger for multi-plant firms (significant at the 10% level only, reflecting their small share of the sample), 1.27 points larger for firms with broad destination portfolios, and 0.858 points larger for firms with broad product portfolios, the latter two both significant at the 1% level. The baseline coefficient falls to 0.0831 and 0.149 in columns (2) and (3) and is no longer distinguishable from zero, so the average effect documented above is concentrated among firms with prior export breadth rather than being uniform across the distribution. The evidence

Table 14: Firm size proxies

	(1)	(2)	(3)
	export$_{fd,t+1}$		
signal$_{fdt}$	0.264**	0.0831	0.149
	(0.105)	(0.104)	(0.106)
signal$_{fdt}$ × multi-plant$_f$	1.11*		
	(0.620)		
signal$_{fdt}$ × broad destinations$_f$		1.27***	
		(0.300)	
signal$_{fdt}$ × broad products$_f$			0.858***
			(0.273)
Observations	934,240	934,240	934,240
R^2	0.819	0.819	0.819
Firm–country FE	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t+1$, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

thus favors the first hypothesis: information is complementary to export capability, lowering the cost of finding a buyer while leaving the fixed costs of serving one in place, so that only firms that have already paid part of those costs can act on what they learn.

7 Firm-level outcomes

Table 15 aggregates the analysis to the firm-year level to ask whether buyer information changed firms' overall export portfolios or merely redirected them. Two treatment measures distinguish the breadth of information from its depth: the number of destinations from which a firm received a signal (*signaled destinations*), and the total number of matched buyers summed across all destinations (*total matched buyers*). Outcomes are the number of export destinations and the number of exported products in the following year; all specifications control for the initial outcome and include firm and year fixed effects, so identification comes from within-firm changes in information exposure over time. Three findings emerge. First, the geographic scope of exporting responds to the depth of information, not its breadth: the count of signaled destinations has no significant effect on next-year destination count (column 1), but the total number of matched buyers does (column 2)—one hundred additional listed buyers raise the destination count by 0.056, relative to a sample mean of roughly 2.5 destinations. This mirrors the dyadic results: what drives entry is the volume of actionable buyer information, not the mere fact of coverage. Second, because total destination counts rise with information, the destination-level entry effects do not reflect pure reallocation across markets: buyer information expanded firms' export portfolios on net rather than

merely redirecting sales from non-signaled to signaled destinations. Third, information breadth is associated with product specialization: firms receiving signals from more destinations reduce their exported product range (column 3), consistent with firms concentrating on product lines for which foreign demand has been identified, while product scope does not respond to buyer depth (column 4). These firm-level results should be read as a complement to, not a substitute for, the dyadic analysis: aggregation to the firm-year level forgoes the within-destination-year comparisons that identify the main results, and the estimates rely on parallel trends across firms with different information exposure. With that caveat, the pattern is coherent: public buyer information led firms to serve more markets with a narrower, demand-matched product range.

Table 15: Firm-level outcomes

	(1)	(2)	(3)	(4)
	dest_count$_{f,t+1}$		product_num$_{f,t+1}$	
signal count$_{ft}$	0.00985		−0.0873***	
	(0.00967)		(0.0318)	
firms sum$_{ft}$/100		0.0516***		0.0726
		(0.0164)		(0.0468)
Observations	26,978	26,978	18,088	18,088
R^2	0.621	0.621	0.714	0.714
Initial outcome	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8 Conclusion

This paper estimates the causal effect of publicly provided information about foreign buyers on firms' export market participation, using the rotating geographic coverage of the Austro–Hungarian *Export-Compass* between 1890 and 1898. Publication raises the probability of entering a destination in the following year by 0.287 percentage points against a baseline of 4.9 percent, a 5.7 percent increase, and signals listing an above-average number of matched buyers raise it by 0.874 points, or 18.3 percent; the effect appears only with a one-year lag, consistent with the October publication date, and firms show no response to listings not yet published. What firms lacked, however, was not information in general but information they could act on. The response is roughly twice the baseline in Mediterranean markets served through the monarchy's own ports and indistinguishable from zero for overseas destinations, and it is three to four times larger when the listed buyers are predominantly wholesale intermediaries—the only signals whose effect persists beyond a year. The same logic appears on the firm side: the average effect is concentrated among firms that already exported to several destinations or across several product lines, and is not distinguishable from zero for firms without prior export breadth. Information lowered the cost of *finding* a buyer while

leaving the fixed costs of *serving* one in place, so that only firms which had already paid part of those costs could act on what they learned. Aggregated to the firm level, it is the number of matched buyers rather than the mere count of covered destinations that raises next-year destination counts, so the estimates reflect net expansion of export portfolios rather than reallocation across markets.

These findings carry a message for the evaluation of export promotion policy. The *Export-Compass* was neither a subsidy nor a logistical intervention but a comparatively cheap information service, and it produced a measurable response—evidence that not knowing *whom* to sell to was a real barrier to trade. The pattern of effects also suggests why randomized evaluations of modern export promotion often find no extensive-margin response (Breinlich et al., 2017; **?**; Ali et al., 2025; **?**): the interventions they study provide general knowledge about the export process, whereas what moved firms here was specific, actionable buyer identities. The complementarity with firm capability qualifies this, however. Buyer information was not a substitute for export capacity, and a service of this kind should not be expected to bring wholly inexperienced firms into foreign markets on its own. Where information is the binding constraint—for firms that can already ship, finance and fulfill an order, but do not know who abroad would place one—public provision appears to have been effective at low cost.

Several limitations bound these conclusions. The outcome is export participation at the destination level rather than export values, so the results cannot be converted into a trade-volume or welfare magnitude without further assumptions. The one-year horizon of the average effect leaves open whether the matches formed were durable, except for the intermediary-heavy signals where persistence is directly observable. Treatment is assigned by matching firms' product descriptions to buyers' reported import categories, and the resulting measurement error attenuates the estimates toward zero. Finally, the exogeneity argument is strongest for the 1896 restart and the timing of re-coverage, where the completion-then-restart rule is explicit, and weaker for the initial 1890 cohort.

Several extensions remain, including within the scope of this project: linking buyer information to firm innovation and product upgrading, and to firm longevity—whether market intelligence fostered durable participation in international trade or merely temporary entry. A quantification of the welfare return to the service, set against its publication cost, would allow the intervention to be compared directly with the subsidy-based instruments that dominate modern export promotion. More broadly, the setting offers what modern administrative data cannot: direct observation of the information itself. Opening this black box further remains a promising avenue for understanding how buyer–seller matches form.

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A Other tables

Table A.1: Effect of buyer-information signals on next-period exporting for firms with destination info for at least 3 years

	(1)	(2)	(3)	(4)
	export$_{t+1}$			
signal$_t$	3.00***	1.27***	0.254**	0.237**
	(0.153)	(0.228)	(0.119)	(0.108)
Mean of dep. var.	.049			
Observations	776,200	776,200	776,200	776,200
R^2	0.001	0.129	0.820	0.841
Firm–country FE	No	No	Yes	Yes
Firm–year FE	No	No	No	Yes
Country–year FE	No	Yes	Yes	Yes

Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.2: Buyer counts as continuous measure

	Dependent variable: Export_{t+1}			
	(1)	(2)	(3)	(4)
Buyers_t (/100)	0.613***		0.987**	
	(0.228)		(0.484)	
log Buyers_t		0.132***		
		(0.0459)		
Max buyers_t (/100)			−0.621	
			(0.699)	
Placebo: Buyers_{t+1} (/100)				−0.337*
				(0.197)
Observations	934,240	934,240	934,240	723,520
R^2	0.819	0.819	0.819	0.912
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

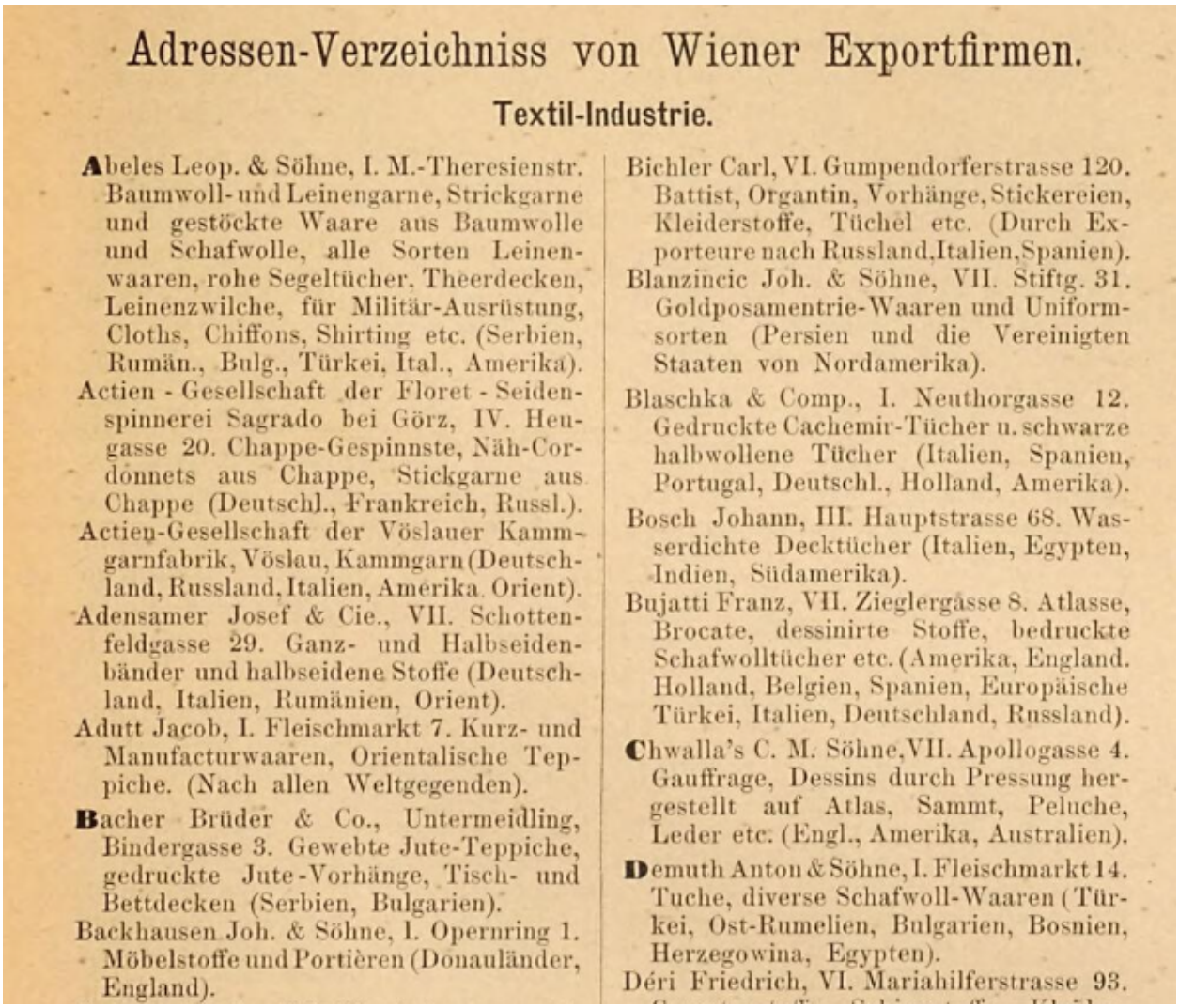
Notes: The dependent variable is an indicator for exporting to destination d in year $t+1$. *Buyers* is the number of product-matched buyers listed in the Compass for destination d in year t, divided by 100; thus, the coefficient measures the change in the probability of exporting associated with an additional 100 listed buyers. Column (2) reports the log version of column (1). Column (3) adds the number of max buyers for the largest product signal. Column (4) replaces column (1) with its one-year lead, i.e. buyers listed in the *following* year's edition, which was not yet available to firms at time t, as a placebo test. Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.3: Buyer counts as continuous measure

	(1)	(2)	(3)	(4)
	Dependent variable: export_{t+1}			
Demand sum (×100)	0.00607***	0.00613***	0.00608***	
	(0.00228)	(0.00228)	(0.00227)	
Merchant share	0.00693***		0.00700***	0.00585**
	(0.00246)		(0.00243)	(0.00246)
Austro-Hungarian share		−0.00183	−0.00253	−0.00525
		(0.00635)	(0.00632)	(0.00633)
Log demand sum				0.00127***
				(0.000464)
Observations	934,240	934,240	934,240	934,240
R-squared	0.819	0.819	0.819	0.819
Firm × country FE	Yes	Yes	Yes	Yes
Firm × year FE	Yes	Yes	Yes	Yes
Country × year FE	Yes	Yes	Yes	Yes

Notes: The dependent variable is an indicator for exporting to destination d in year $t+1$. *Buyers* is the number of product-matched buyers listed in the Compass for destination d in year t, divided by 100; thus, the coefficient measures the change in the probability of exporting associated with an additional 100 listed buyers. Column (2) reports the log version of column (1). Column (3) adds the number of max buyers for the largest product signal. Column (4) replaces column (1) with its one-year lead, i.e. buyers listed in the *following* year's edition, which was not yet available to firms at time t, as a placebo test. Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B Other figures



Adressen-Verzeichniss von Wiener Exportfirmen.

Textil-Industrie.

Abeles Leop. & Söhne, I. M.-Theresienstr. Baumwoll- und Leinengarne, Strickgarne und gestöckte Waare aus Baumwolle und Schafwolle, alle Sorten Leinenwaaren, rohe Segeltücher. Theerdecken, Leinenzwilche, für Militär-Ausrüstung, Cloths, Chiffons, Shirting etc. (Serbien, Rumän., Bulg., Türkei, Ital., Amerika).

Actien - Gesellschaft der Floret - Seidenspinnerei Sagrado bei Görz, IV. Heugasse 20. Chappe-Gespinnste, Näh-Cordonnets aus Chappe, Stickgarne aus Chappe (Deutschl., Frankreich, Russl.).

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Adensamer Josef & Cie., VII. Schottenfeldgasse 29. Ganz- und Halbseidenbänder und halbseidene Stoffe (Deutschland, Italien, Rumänien, Orient).

Adutt Jacob, I. Fleischmarkt 7. Kurz- und Manufacturwaaren, Orientalische Teppiche. (Nach allen Weltgegenden).

Bacher Brüder & Co., Untermeidling, Bindergasse 3. Gewebte Jute-Teppiche, gedruckte Jute-Vorhänge, Tisch- und Bettdecken (Serbien, Bulgarien).

Backhausen Joh. & Söhne, I. Opernring 1. Möbelstoffe und Portièren (Donauländer, England).

Bichler Carl, VI. Gumpendorferstrasse 120. Battist, Organtin, Vorhänge, Stickereien, Kleiderstoffe, Tüchel etc. (Durch Exporteure nach Russland, Italien, Spanien).

Blanzincic Joh. & Söhne, VII. Stiftg. 31. Goldposamentrie-Waaren und Uniformsorten (Persien und die Vereinigten Staaten von Nordamerika).

Blaschka & Comp., I. Neuthorgasse 12. Gedruckte Cachemir-Tücher u. schwarze halbwollene Tücher (Italien, Spanien, Portugal, Deutschl., Holland, Amerika).

Bosch Johann, III. Hauptstrasse 68. Wasserdichte Decktücher (Italien, Egypten, Indien, Südamerika).

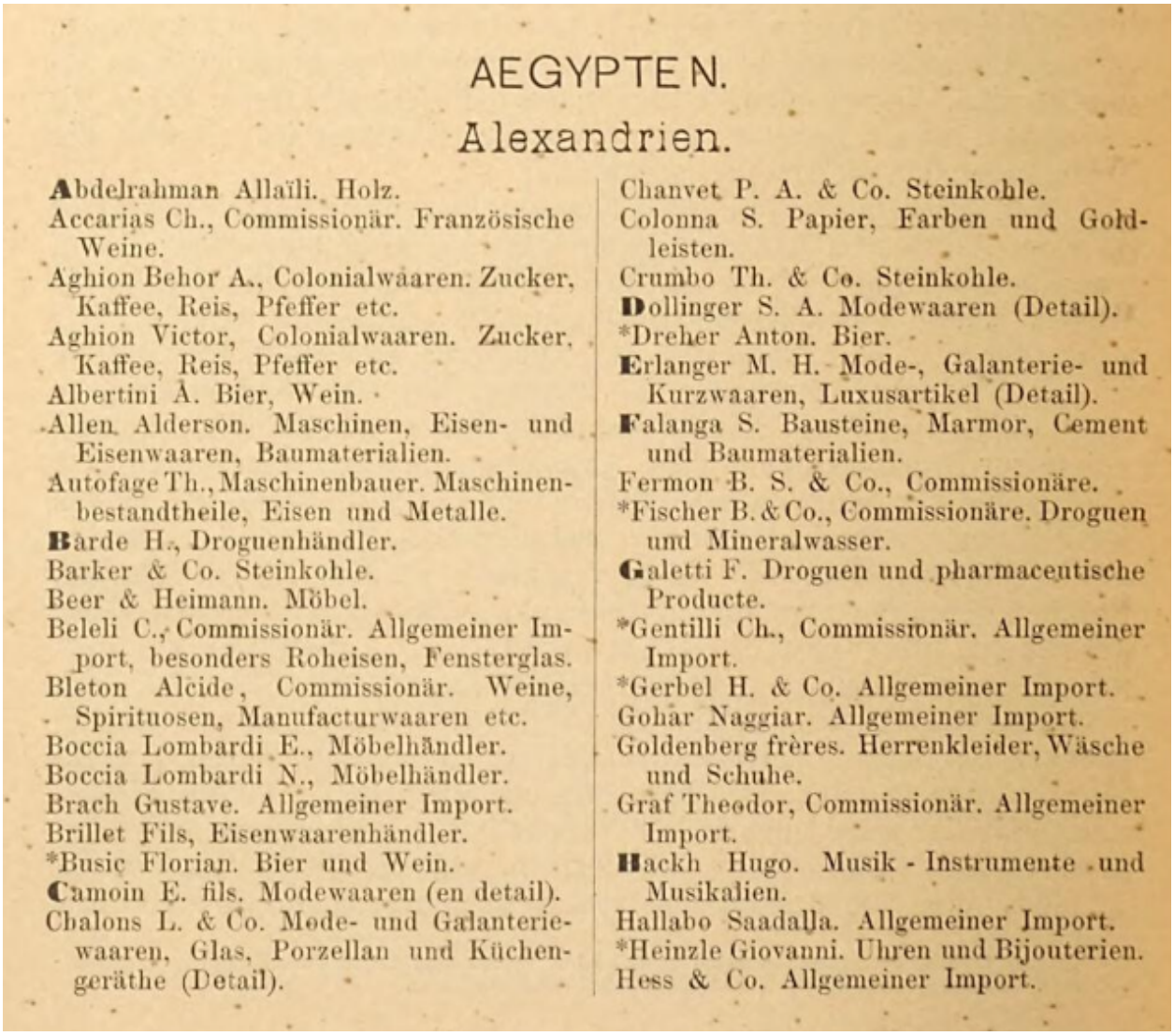
Bujatti Franz, VII. Zieglergasse 8. Atlasse, Brocate, dessinirte Stoffe, bedruckte Schafwolltücher etc. (Amerika, England. Holland, Belgien, Spanien, Europäische Türkei, Italien, Deutschland, Russland).

Chwalla's C. M. Söhne, VII. Apollogasse 4. Gauffrage, Dessins durch Pressung hergestellt auf Atlas, Sammt, Peluche, Leder etc. (Engl., Amerika, Australien).

Demuth Anton & Söhne, I. Fleischmarkt 14. Tuche, diverse Schafwoll-Waaren (Türkei, Ost-Rumelien, Bulgarien, Bosnien, Herzegowina, Egypten).

Déri Friedrich, VI. Mariahilferstrasse 93.

Figure B.1: Example of raw data from exporting firms



AEGYPTEN.

Alexandrien.

Abdelrahman Allaïli. Holz.
Accarias Ch., Commissionär. Französische Weine.
Aghion Behor A., Colonialwaaren. Zucker, Kaffee, Reis, Pfeffer etc.
Aghion Victor, Colonialwaaren. Zucker, Kaffee, Reis, Pfeffer etc.
Albertini A. Bier, Wein.
Allen Alderson. Maschinen, Eisen- und Eisenwaaren, Baumaterialien.
Autofage Th., Maschinenbauer. Maschinenbestandtheile, Eisen und Metalle.
Barde H., Droguenhändler.
Barker & Co. Steinkohle.
Beer & Heimann. Möbel.
Beleli C., Commissionär. Allgemeiner Import, besonders Roheisen, Fensterglas.
Bleton Alcide, Commissionär. Weine, Spirituosen, Manufacturwaaren etc.
Boccia Lombardi E., Möbelhändler.
Boccia Lombardi N., Möbelhändler.
Brach Gustave. Allgemeiner Import.
Brillet Fils, Eisenwaarenhändler.
*Busic Florian. Bier und Wein.
Camoin E. fils. Modewaaren (en detail).
Chalons L. & Co. Mode- und Galanteriewaaren, Glas, Porzellan und Küchengeräthe (Detail).
Chanvet P. A. & Co. Steinkohle.
Colonna S. Papier, Farben und Goldleisten.
Crumbo Th. & Co. Steinkohle.
Dollinger S. A. Modewaaren (Detail).
*Dreher Anton. Bier.
Erlanger M. H. Mode-, Galanterie- und Kurzwaaren, Luxusartikel (Detail).
Falanga S. Bausteine, Marmor, Cement und Baumaterialien.
Fermon B. S. & Co., Commissionäre.
*Fischer B. & Co., Commissionäre. Droguen und Mineralwasser.
Galetti F. Droguen und pharmaceutische Producte.
*Gentilli Ch., Commissionär. Allgemeiner Import.
*Gerbel H. & Co. Allgemeiner Import.
Gohar Naggiar. Allgemeiner Import.
Goldenberg frères. Herrenkleider, Wäsche und Schuhe.
Graf Theodor, Commissionär. Allgemeiner Import.
Hackh Hugo. Musik - Instrumente und Musikalien.
Hallabo Saadalla. Allgemeiner Import.
*Heinzle Giovanni. Uhren und Bijouterien.
Hess & Co. Allgemeiner Import.

Figure B.2: Example of raw data from importing firms



Figure B.3: Matching importers and exporters through relevant products

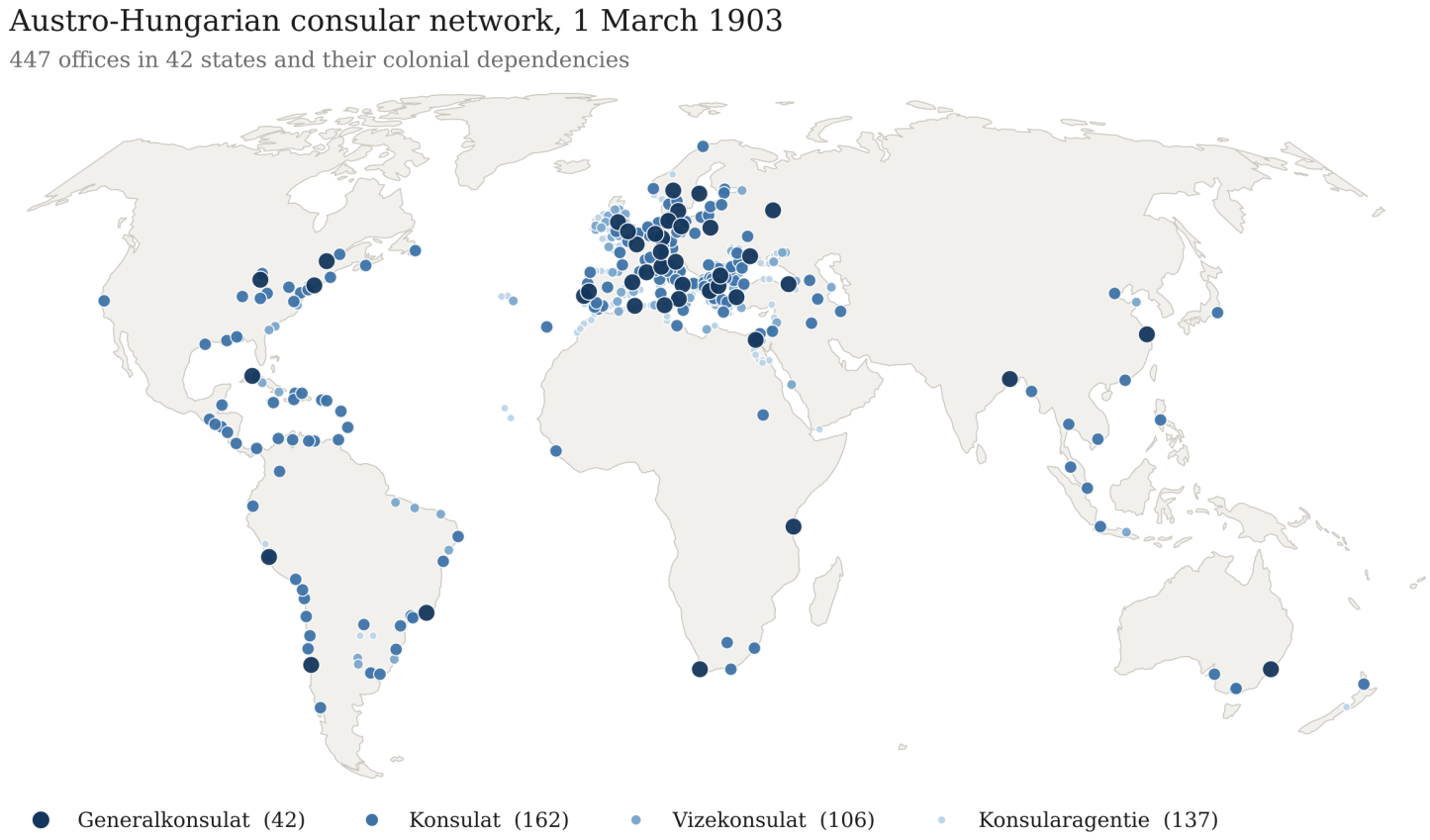
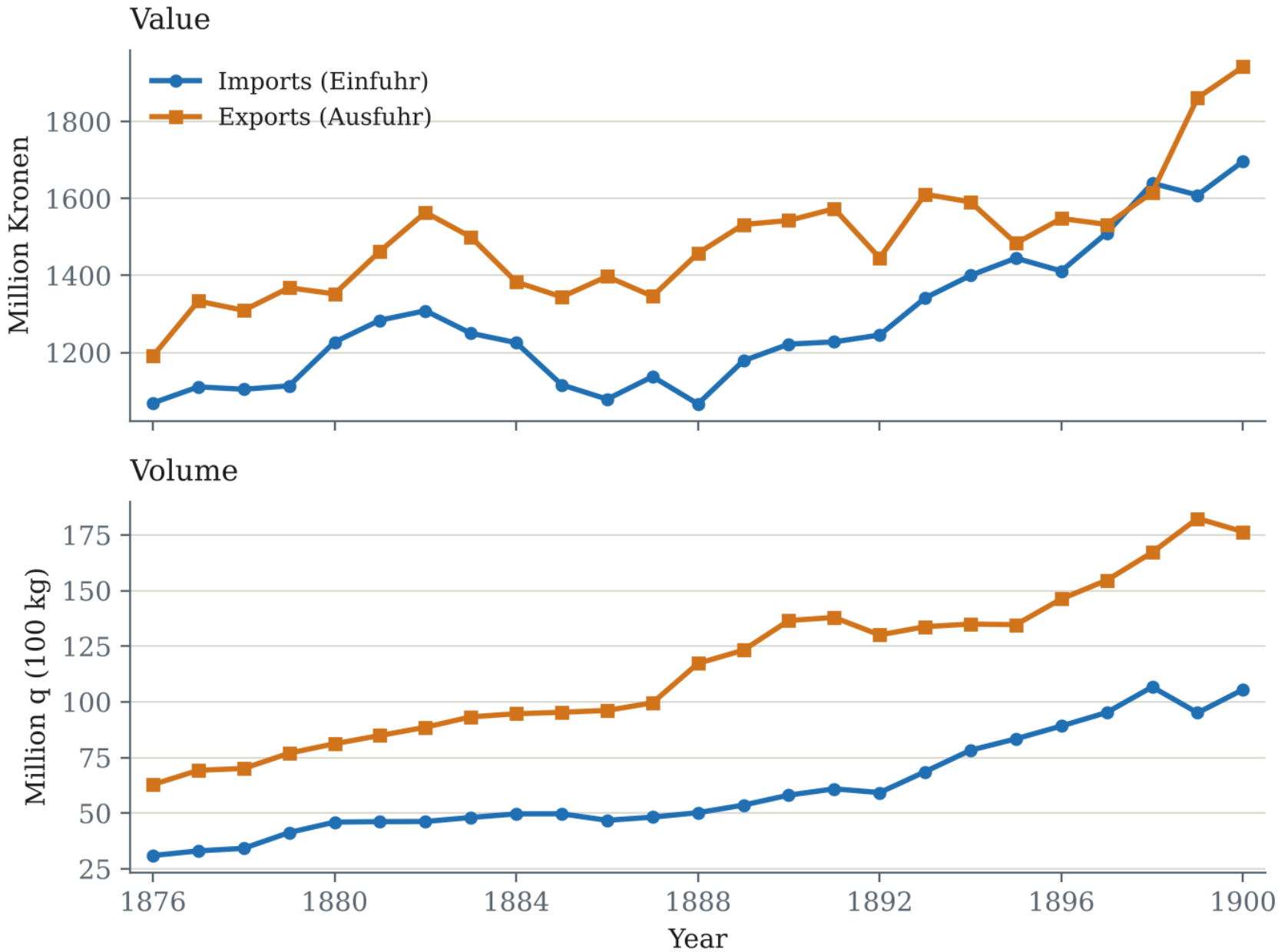


Figure B.4: Consular offices worldwide

Austria-Hungary: special trade, 1876–1900



C Data Construction

C.1 Source Data

The underlying data source is the *Austro-Hungarian Export Compass* (*Österreichisch-Ungarischer Export-Compass*), a commercial directory published annually listing exporting firms across the Habsburg Monarchy. I use the editions from 1890 to 1898. The raw data was digitized into Excel spreadsheets, with each row containing a single firm entry as a free-text string. These entries typically combine the firm's name, location or address, product descriptions, and miscellaneous information (e.g., legal form, advertisements, agency roles) in a single unstructured line of German-language text.

The data construction pipeline consists of six steps, summarized in Figure C.1 and described in detail below.

Figure C.1: Data Construction Pipeline

Step	Description
1	Parsing raw firm entries into structured fields
2	Product category classification
3	Export destination standardization
4	Cross-year firm matching
5	Panel assembly and firm characteristic merge
6	Firm–country–year long panel and import demand matching

C.2 Step 1: Parsing Firm Entries into Structured Fields

Given the heterogeneous and historically idiosyncratic formatting of the entries, manual rule-based parsing proved infeasible at scale. I employ a large language model (LLM)—specifically OpenAI's GPT-5.1—via the OpenAI API to decompose each unstructured firm entry into four structured fields:

1. **Name**: the firm name, including legal form (e.g., "& Comp.", "A.G.", "Gebrüder").
2. **Address**: the location, street, district, or regional designator (e.g., "Bez. Krumau", "Mähren").
3. **Product**: the goods or product categories associated with the firm.
4. **Other information**: residual content such as references to advertisements, speciality notes, or agency relationships.

An additional binary flag (`parse_ok`) indicates whether the entry was successfully recognized as a genuine firm listing, as opposed to an advertisement or editorial comment.

Each raw text entry is first cleaned by stripping whitespace and collapsing multiple spaces. Entries are then grouped into batches of 40 and submitted to the LLM in a single API call. The model receives a detailed German-language system prompt instructing it to separate each entry into the four target fields, with explicit rules about field assignment—for instance, that geographic qualifiers belong in the address field rather than the product field, and that product information can

be inferred from the firm type when not explicitly stated (e.g., a "Zuckerfabrik" produces sugar). The API temperature parameter is set to 0, ensuring deterministic output for reproducibility.

Each unique input text is hashed (SHA-1) and the corresponding parsed result is stored in a persistent JSON cache. This avoids redundant API calls when re-running the pipeline and provides robustness against interruptions. Failed API calls are retried up to three times with exponential back-off.

C.3 Step 2: Product Category Classification

The product descriptions extracted in Step 1 are free-text strings that frequently list multiple goods separated by commas, semicolons, or the conjunction "und" (e.g., "Colonialwaaren, Eisen und Glas"). A second pipeline assigns each firm one or more standardized product categories drawn from a controlled German-language vocabulary.

Each product string is first split into individual tokens using commas, semicolons, slashes, ampersands, and the word "und" as delimiters. All unique tokens across the dataset are collected, and those not yet cached are submitted in batches of 20 to GPT-5.1. The model receives the current list of existing categories and is instructed to assign each token to exactly one category, creating a new category if no suitable one exists. The prompt explicitly prohibits catch-all labels such as "Sonstiges" (miscellaneous) and encourages fine-grained distinctions—for instance, distinguishing "Baumwollwaren" (cotton goods) from "Seidenwaren" (silk goods) rather than grouping them under a generic "Textilien" label. Similarly, colonial goods are disaggregated into specific commodities (coffee, sugar, tea, tobacco, etc.).

A deterministic post-processing step merges known near-synonyms arising from historical spelling variation. For example, "Colonialwaaren" and "Kolonialwaren" are unified, as are "Eisenwaaren" and "Eisenwaren," reflecting the orthographic reform from *aa* to *a* that occurred in this period. The category vocabulary is initialized with a seed set of approximately 30 categories spanning major sectors (beverages, foodstuffs, textiles, metals, machinery, etc.) and grows organically as the model encounters new product types. As in Step 1, all token-to-category mappings are cached for reproducibility.

C.4 Step 3: Export Destination Standardization

The Export Compass records not only the goods each firm traded but also its export destinations, recorded as free-text strings listing countries, regions, and abbreviated place names in the conventions of the 1890s (e.g., "Ital.", "Rumän.", "Orient"). A third pipeline standardizes these entries into a controlled vocabulary of country and region names.

Each destination string is split on commas, semicolons, slashes, dashes, and the conjunction "und" into individual tokens. Stopwords and noise tokens (e.g., "etc.", "via", "über") are discarded. Before invoking the LLM, a fast deterministic lookup resolves common abbreviations and spelling variants (e.g., "Ital." → "Italien", "Holland" → "Niederlande", "England" → "Großbritannien"). Tokens that match an existing category after Unicode normalization and case-folding are also resolved locally, avoiding unnecessary API calls.

Remaining unresolved tokens are submitted in batches of 10 to GPT-5.1. The model receives the current list of standardized categories and is instructed to map each input to exactly one existing category or, if none fits, to create a new one. The prompt explicitly forbids composite categories (e.g., "Deutschland und Schweiz") and catch-all labels, and instructs the model to stay as close as possible to the literal text—mapping "Bulgarien" to "Bulgarien" rather than abstracting it to "Balkan."

The category list is initialized with a curated set of approximately 60 entries covering the major countries and macro-regions relevant to late-nineteenth-century Austro-Hungarian trade (e.g., European states, Ottoman territories, colonial destinations, and broad regional labels such as "Orient," "Levante," and "Überseeisch"). After standardization, each firm's destination list is split into two sub-lists: specific countries (e.g., "Serbien," "Rumänien") and broad regions (e.g., "Balkan," "Südamerika," "Weltweit"), based on a predefined set of region labels.

C.5 Step 4: Cross-Year Firm Matching

To track firms over time and construct a panel dataset, firm entries are linked across all available years (1890–1898) using a combination of exact and fuzzy name matching.

Firm names are preprocessed by stripping diacritics, lowercasing, removing punctuation, expanding abbreviations (e.g., "&" → "und"), and stripping legal suffixes and corporate designators (e.g., "A.G.", "Ges.m.b.H.", "Gebrüder", "& Co.", "Fabrik"). To reduce the computational burden, firms are grouped into blocks defined by the first two characters of the normalized name. Candidate matches are drawn primarily from the same block; if no suitable candidate is found, the full pool is searched as a fallback.

For each pair of years, firms whose normalized names are identical are matched first (exact match, score = 100). Remaining unmatched firms are compared using the token-set-ratio metric from the RapidFuzz library, which is robust to word reordering and partial overlap. A match is accepted only if the similarity score meets or exceeds a threshold of 90. A match between firm i in year t and firm j in year t' is accepted only if j is i's highest-scoring candidate *and* i is j's highest-scoring candidate—a mutual (symmetric) best-match criterion that guards against false positives.

All accepted pairwise links across all $\binom{9}{2} = 36$ year-pairs are fed into a Union-Find (disjoint-set) data structure. This groups firm observations that are transitively linked—if firm i in 1890 matches firm j in 1892, and j matches firm k in 1894, all three are assigned to the same cluster. Each cluster receives a stable panel identifier (`ex_uid`).

C.6 Step 5: Panel Assembly and Firm Characteristic Merge

The pairwise match edges from Step 4 are represented as an undirected graph using the NetworkX library, where each node is a year–firm pair and each edge is an accepted match link. Connected components identify firm clusters, each assigned a stable panel identifier.

The graph is converted into a long-format panel with one row per firm-year observation. A wide-format version is also produced, with one row per firm cluster and separate columns for the firm's original ID in each year. The long panel is then merged with the enriched yearly firm files produced by Steps 1–3, attaching the full set of firm-level characteristics—parsed name, address, product descriptions, standardized product categories, export destinations, and standardized country and region classifications—to each panel observation.

C.7 Step 6: Firm–Country–Year Panel and Import Demand Matching

The final analytical dataset is constructed by reshaping the panel into a firm–country–year structure and merging it with external data on import demand.

A master list of all destination countries appearing anywhere in the data is compiled. Each firm-year observation is crossed with the full set of countries, producing a row for every possible firm–country–year combination. A binary indicator is set to 1 if the firm lists that country among its export destinations in that year, and 0 otherwise. This structure enables gravity-style estimation of both the intensive and extensive margins of trade.

To measure the demand environment each firm faces in each destination, the dataset is further exploded along the product dimension: each firm–country–year row is expanded into firm–country–year–category rows using the standardized product categories from Step 2. These are then merged with an external file recording the number of importing firms by country, year, and product category—a proxy for destination-specific import demand. The product-level demand matches are aggregated back to the firm–country–year level, producing summary measures: the sum, mean, and maximum of matched import demand across the firm's product categories, as well as the count of categories for which a demand match was found.

The resulting dataset—one row per firm–country–year with export indicators, firm characteristics, and demand measures—is exported both as an Excel file and as a Stata `.dta` file.

C.8 Validation

Entries flagged with `parse_ok = false` by the model in Step 1 were reviewed manually. A random sample of successfully parsed entries was also inspected to verify the quality of field separation. For the product categorization in Step 2, category frequency distributions were examined to detect overly broad or inconsistent labels. In Step 4, match quality was assessed through the distribution of fuzzy match scores and the share of exact versus fuzzy matches across year-pairs. The deterministic (zero-temperature) setting and the caching mechanism in Steps 1–3 ensure that results are reproducible across runs.

D Descriptive statistics from entity resolution

D.1 Data Construction and Firm Matching

The dataset is constructed by digitising the Austro-Hungarian Export Compass for the years 1890–1898. Table D.1 reports the number of firm entries per edition after removing within-year duplicates. The raw entry count grows steadily from 3,255 in 1890 to 11,355 in 1898, reflecting both genuine market expansion and the increasing coverage of later editions. After deduplication the usable sample rises from 3,173 to 10,320 firm-year observations; duplicate rates range from 2.5% in 1890 to 9.1% in 1898.

Table D.1: Sample size by year

Year	Raw entries	After deduplication	Duplicates removed
1890	3,255	3,173	82
1891	5,533	5,331	202
1892	6,172	5,907	265
1893	7,679	7,061	618
1894	8,989	8,114	875
1895	9,605	8,727	878
1896	10,266	9,418	848
1897	11,025	10,074	951
1898	11,355	10,320	1,035

D.2 Cross-Year Firm Matching

To track firms across editions, each firm name in year t is matched against all firm names in every other year t' using a combination of exact string matching and fuzzy matching based on edit-distance scores normalised to $[0, 100]$. Across all 36 year pairs, a total of 151,708 cross-year matches are identified, of which 132,854 (87.6%) are exact and 18,854 (12.4%) are fuzzy. The mean match score is 99.36 and the median is 100, indicating that the vast majority of fuzzy matches involve only minor name variations (e.g. spelling differences or abbreviations) rather than substantive discrepancies.

Table D.2 reports match counts and quality for consecutive year pairs. The number of matched firms is naturally highest for adjacent years and declines as the gap widens, consistent with firm entry and exit over time. Match quality remains high throughout: mean scores for consecutive pairs all exceed 99.6.

D.3 Firm Persistence Across Editions

Table **??** reports the distribution of firms by the number of editions in which they are observed. The majority of firms—29.7%—appear in only a single edition, suggesting substantial turnover in the exporter population. At the other extreme, 1,513 firms (8.1%) are observed in all nine editions, forming a stable core of persistent exporters. Firms observed in two or three editions together account for a further 28.9% of the sample.

Table D.2: Cross-year matches for consecutive year pairs

Year pair	Total matches	Exact	Fuzzy	Mean score
1890–1891	2,789	2,587	202	99.64
1891–1892	4,248	3,974	274	99.66
1892–1893	4,404	4,033	371	99.54
1893–1894	5,548	4,939	609	99.51
1894–1895	6,222	5,761	461	99.61
1895–1896	6,742	6,066	676	99.48
1896–1897	7,200	6,462	738	99.52
1897–1898	7,525	7,012	513	99.69
All 36 pairs combined				
All pairs	151,708	132,854	18,854	99.36

E Consular market reports

E.1 Source

The signal measure is built from the *Rathschläge der k. und k. Consular-Behörden*, a section of the annual *Export-Compass: commerzielles Jahrbuch für die Interessenten des österreichisch-ungarischen Ausfuhr-Handels* (Vienna; Österreichische Nationalbibliothek, shelfmark 394928-B, catalogue record AC02742619). The same yearbook carries the buyer registry used elsewhere in the paper, so signals and buyer lists are drawn from a single publication and reach the same readership.

Each entry in the *Specielle Artikel* subsection is a short assessment by a named consular post of a named good in that post's market: whether Austro-Hungarian goods are selling there, what is holding them back, and which competitor countries are present. The unit is therefore post × good × edition.

Three features of the source govern what can be measured.

Vintage varies within a volume. The volume preambles state that volume N carries reports received between July of $N-1$ and June of N, covering report years $N-2$ and $N-1$. From the 1894 edition onward each entry is headed with its own report year (*Mailand (pro 1892).*), and these stamps confirm the rule directly: the 1894 volume carries 21 entries from 1892 reports, 94 from 1893 and one from 1894. Report year is consequently an entry-level variable, not a volume-level one, and consecutive volumes overlap by one report year.

Only new observations were printed. The 1895 preamble states that a post's report was printed only where it contained something new. The absence of a signal for a post–good–year therefore means "the consul had nothing new to add," not "the market was neutral." This is conceded in Section E.6 rather than argued away.

Two volumes are double-year and one year is missing. The digitised run is 1888–1895, 1897/98 and 1898/99; there is no 1896 edition in the ÖNB holdings list for this record and none surfaces in the trade, so 1896 should be treated as a gap in the source rather than in the digitisation.

E.2 Extraction

The scans carry an OCR text layer, but the structure of the page is carried by typography rather than by punctuation: a bold paragraph opener marks the consular post, a letterspaced run marks the good, a centred heading marks the section, and a paragraph indent marks the entry boundary. OCR collapses letterspacing (`M ö b e l` becomes `Möbel`), so a model reading raw OCR must guess which noun is the product — an error that is silently correlated with the content of the text. The pipeline therefore recovers typography from page geometry and uses a language model only where typography runs out (translation of product names, competitor extraction).

The printed schema drifts across editions. The 1890 and 1891 volumes head sections with an industry group and identify the good by letterspacing; from 1892 the heading is the good itself. Schema is detected per volume by the presence of a goods subtitle, which a group heading always carries and a product heading never does. Because the printed headings are not stable across editions — 1891 merges textiles and clothing, 1892 uses about thirty alphabetical goods, 1893 forty-eight and 1894 seventy-five — all headings are crosswalked onto a common twelve-group industry taxonomy, and pooling is done on that taxonomy rather than on the printed heading.

Two integrity checks are run on every new edition. Non-entry material (the Chambers of Commerce name lists at the end of the 1893 excerpt) is rejected only when two independent signals agree: the post fails the gazetteer *and* the capitalised-token fraction is at least 0.65, against 0.18–0.37 for report prose and 0.73–0.97 for name lists. Reprints are detected by same-post cross-edition

text similarity; at a Jaccard threshold of 0.30 the 1890–94 corpus contains two, both Algiers 1891–1892.

The resulting corpus is reported in Table E.1. Entries from 1895–97 are extracted from prose rather than from a printer's product designation and are flagged as a distinct measurement; the drop in counts after 1894 reflects the change in extraction, not a decline in consular reporting.

E.3 Coding the signal

Two assessments are latent in a typical entry, and they are frequently *inversely* related within the same sentence. Of Viennese window glass in Sofia the consul writes that Austro-Hungarian goods are absent, *but* that "so wäre es möglich ... hier durchzudringen"; of matches in Smyrna, that Austro-Hungarian product "beherrscht noch immer den Markt, *doch* ... könnte der Markt verloren gehen." A single favourability score orders these two cases wrongly.

The paper therefore codes the dimension a reading firm would act on. Let $opportunity_{pgt} \in \{\text{encouraging}, \text{neutral}, \text{discouraging}\}$ record whether the consul urges entry into or warns away from market p for good g. Detection is grammatical rather than sentiment-based: the German subjunctive II (*wäre, könnte, dürfte, sollte, fänden*) is the mood in which an attainable gain is asserted, and its presence in a recommendation clause is the syntactic marker of an encouraging signal.

The second dimension — the current position of Austro-Hungarian goods in the market — is not used. Against a hand-coded gold set it is recovered at only 47 per cent accuracy, and it mechanically tracks existing trade, so a balance test on it approaches the tautology "is trade where trade is." Table E.4 reports agreement rates: *opportunity* is recovered at 80 per cent three-way and 87 per cent binary.

Roughly 13 per cent misclassification attenuates every coefficient below toward zero. This works *against* the conditional null in Section E.5, which is harder to defend under attenuation, and *for* the symmetry result, where a true asymmetry would be harder to detect.

E.4 Matching signals to buyer cells

Signals are observed at post × good × edition and must be matched to the buyer panel at country × category × year. Posts are resolved to countries through a consular gazetteer; goods are matched to the 339-category buyer taxonomy by a four-pass matcher (exact, stem, substring, manual) that records the match type used. Table E.2 reports the funnel.

Two features of the attrition matter for interpretation. First, the unmatched 29 per cent are not random: foodstuffs and beverages (*Butter, Eier, Cognac, Champagner, Cement*) have no counterpart in the buyer taxonomy, so the matched sample tilts toward manufactures and the claims below should be read as claims about the manufacturing part of the sample. Second, the matched signals cover 65 of 3,141 buyer cells, or 2.1 per cent. These are excerpts of 6–22 pages from volumes running 542–826 pages; extracting the full *Specielle Artikel* sections would multiply the intensive margin several-fold and is the binding constraint on precision.

At the firm level three construction choices follow from this. Coverage ($covered_{ct}$, equal to one if any category in country c carries a report in year t) is merged on country × year rather than aggregated through the category explode, since it is a property of the country-year; only 10 of 47 country-years in the buyer panel are covered. Signal counts summed over a firm's categories are mechanically increasing in product breadth, so the share of a firm's matched categories carrying a report is used in place of the raw sum, or else the raw count is entered with the number of matched categories on the right-hand side. Outside covered country-years the signal variables are set to

missing rather than to zero: there, zero records that nothing was extracted for that destination, and a regression on the unmasked panel compares coverage rather than signal.

E.5 Is consular attention balanced with respect to the buyer list?

The identification concern is that the consular signal and the buyer list are placed in the same product-markets: if the consul writes favourably precisely where many buyers are listed, an export response to a listing could reflect the market opportunity rather than the information. Table E.3 tests this on two margins, with opposite results.

On the *extensive* margin — whether a product-market is discussed at all — there is a clear relationship with buyer-list size: a log point more listed firms raises the probability of any signal by 3.74 percentage points (s.e. 0.0066) against a base rate of 6.2 per cent. But the relationship is symmetric across tones. Encouraging and discouraging signals load almost identically (+1.01 and +1.02 percentage points), and the difference in slopes is −0.0003 (s.e. 0.0056, $p = 0.95$, 95 per cent CI $[-0.011, +0.011]$). Complaints sit in slightly *larger* cells than praise: 39.5 listed firms on average against 33.9, with 15.7 where no signal is reported.

On the *intensive* margin — conditional on a product-market being discussed, what the consul says about it — there is no relationship. The probability that a reported signal is encouraging falls by 4.3 percentage points per log firm (s.e. 0.0514, $p = 0.40$, CI $[-0.144, +0.058]$) against a base of 36.9 per cent; the share of encouraging entries gives the same answer.

Together: the probability of being reported on at all rises with buyer-list size, but symmetrically across encouraging and discouraging reports, so consular attention does not select on tone; and conditional on being reported on, the tone is uncorrelated with the number of listed buyers. The claim the design needs is the conditional null, and the symmetry result is what prevents the extensive margin from contaminating it. The stronger claim — that buyer information is uncorrelated with positive signals full stop — is *not* supported and is not made.

Two specification notes. The regression is of the probability of an encouraging signal on log listed firms rather than the reverse, so that the confidence interval is in probability units against an interpretable base; the reversed form yields $[-0.92, +0.37]$ in log points and is uninformative. The sample is restricted to country-years the consular corpus covers, since comparing a cell in a covered country-year with one in an uncovered country-year compares coverage rather than signal. Country fixed effects are partly collinear with year — several countries appear in a single edition — so the year-fixed-effect rows are primary.

E.6 Limitations

Selection on absence. Because only new observations were printed, orthogonality among observed cells does not establish orthogonality in the full sample. The balance test speaks to what the consuls said, not to what they chose to omit.

Partial coverage of the volume. The *Specielle Artikel* section is not the whole of a volume's consular content. In 1890 the companion *Allgemeine Bemerkungen* section covers 41 posts and *Specielle Artikel* 33, with only 13 in both — and the two sets are not random with respect to the outcome, since a post with something to say about a specific good appears in one section and a post with something to say about market conditions in the other. Any measure of "did post p report in year t" built from *Specielle Artikel* alone is therefore incomplete, and in a way correlated with the type of information reported. Appendix A describes what the second section contains.

Measurement change at 1895. Entries from 1895 onward are inferred from prose rather than taken from the printer's designation.

Table E.1: The consular report corpus, 1890–1897

	Tone					
Edition	Encour.	Discour.	Neutral	Entries	Posts	Section
1890				121		Spec. Artikel
1891				24		Spec. Artikel
1892				43		Spec. Artikel
1893				67		Spec. Artikel
1894				116		Spec. Artikel
1895				18		Allg. Bemerk.
1896				28		Allg. Bemerk.
1897				9		Allg. Bemerk.
Total	224	101	101	426	79	

237 distinct product designations; median entry 59 words, maximum 466. Entry-level report year is populated for 162 of 426 entries (stamped from the 1894 edition onward). Per-edition tone splits are in `tables/desc_corpus.tex`, generated by `src/descriptives.py`.

Coverage. 2.1 per cent of buyer cells carry a matched signal, and 10 of 47 country-years are covered at all. Results are correspondingly imprecise on the intensive margin.

A The *Allgemeine Bemerkungen* section: mechanisms

Each *Export-Compass* volume carries, alongside the good-level *Specielle Artikel*, a section of *Allgemeine Bemerkungen* in which each post comments on the conditions of doing business in its market. The unit is post × year rather than post × good × year, and the content is not about outcomes in particular goods but about the frictions that produce them. Table A.1 reports theme prevalence across the 42 posts reporting in 1890.

The three most common subjects map directly onto the information-friction channels this paper studies.

The consulate as an information intermediary, with observable volume. The Philippopel post reports that the number of inquiries it received about businesspeople resident in its district rose by almost half against the previous year, and that it obtains credit information through local bank branches; the London post offers to procure credit reports through accredited information bureaux. This is a measurable demand for information, reported by the same office year on year.

Intermediary presence as the competitive margin. The Piraeus–Athens post attributes German success not to price or quality but to "a number of excellent German commission agents, in whose hands the entire German import business rests"; the Prevesa post asks for a resident commission agent carrying samples.

Contracting frictions. The Nish post sets out clause by clause how Serbian bill-of-exchange law differs from Austrian — an acceptance requires the word *Angenommen* in the acceptor's own hand, and a wife's signature is invalid without her husband's consent. Livorno and Smyrna report fraudulent bankruptcies and the futility of legal action for small sums. Rhodes notes that German is unknown locally and correspondence should be in French or Italian.

Table E.2: From consular entries to buyer cells

	Entries
Extracted, 1890–94 (*Specielle Artikel*)	372
Plus 1895–97 product lead-ins	426
Post resolved to country	403
Good matched to buyer category	301
Both	282
Distinct buyer cells with a signal	65
Buyer cells, all country-years	3,141
Buyer cells, covered country-years	1,046

Match type: exact 142, stem 117, substring 31, manual 11, unmatched 125. Covered country-years are 10 of 47, spanning 9 countries; 6.2 per cent of cells in covered country-years carry a signal.

Table E.3: Balance: buyer-list size and the consular signal

Dependent variable	Coef.	S.e.	95% CI	Base
Panel A. Extensive margin ($n = 1{,}046$ cells)				
Any signal	0.0374	0.0066	[0.024, 0.050]	0.062
Encouraging signal	0.0101	0.0040	[0.002, 0.018]	0.023
Discouraging signal	0.0102	0.0036	[0.003, 0.017]	0.017
Difference in slopes	−0.0003	0.0056	[−0.011, 0.011]	
Panel B. Intensive margin ($n = 65$ cells)				
Pr(encouraging | discussed)	−0.0430	0.0514	[−0.144, 0.058]	0.369
Share encouraging	−0.0444	0.0516	[−0.145, 0.057]	0.369

Each row is a separate regression of the stated outcome on the log number of listed buyers in the cell, with year fixed effects, restricted to country-years covered by the consular corpus. Mean listed firms: 15.7 where no signal is reported, 33.9 encouraging, 39.5 discouraging. Difference in slopes $p = 0.95$. Country fixed effects are rank-deficient against year and are reported in the appendix only. Full output in `tables/balance_tests.csv`.

Table E.4: Coder validation against a hand-coded gold set

Dimension	Three-way	Binary
opportunity (used)	0.80	0.87
Austro-Hungarian position (not used)	0.47	—

$n = 30$ entries. Off-diagonal errors on *opportunity*: three encouraging coded discouraging, two discouraging coded neutral, one neutral coded encouraging.

Table A.1: Themes in *Allgemeine Bemerkungen*, 1890

Theme	Posts
Information intermediaries (agents, commission houses, travelling salesmen, samples, consulate inquiries)	25
Price competition	22
Credit and default risk	21
Named competitor countries	20
Shipping and direct steamer links	14
Law and institutions	14
Packaging and quality complaints	12
Language of correspondence	9
Currency and exchange	7
Tariffs and customs	6
Labour and emigration	3

Of 42 posts reporting in the 1890 *Allgemeine Bemerkungen* section. Themes are multiple-label, so columns do not sum to 42. Named competitors are Germany most often, then Italy, England and Japan. Entry length ranges from 227 to 4,900 characters.

The section is currently extracted only for 1890, and from an OCR'd text file rather than from the scan, so the typographic signals the main pipeline relies on are absent and only the short-line-then-paragraph structure survives. The themes are not mutually exclusive — the Sofia entry covers currency, credit and intermediaries at once — so they are coded as multiple labels rather than as a classification.